

Dramatic Shifts Over Land in CMIP6

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1 Introduction

2 Data & Methodology

3 Results

Below, we list the components of the land-based climate system that undergo dramatic shifts. These shifts can be classified as either gradual or abrupt transitions, depending on how they evolve. Regardless of the type, we refer to them in this text as strong non-linear surprises (SNSs).

3.1 Amazonian basin

We identified four cases in which parts of northern South America, including the Amazon basin, undergo dramatic shifts in vegetation characteristics. Below, we describe these shifts and provide possible explanations for the mechanisms driving such transitions. The first three cases are based on dynamic vegetation models, in which vegetation types (e.g., trees, grasses, shrubs) can compete with one another. The fourth case, however, comes from a non-dynamic vegetation model, where vegetation types cannot evolve naturally but the leaf-area index can still respond to climatic perturbations. Finally, at the end of this section, we report three climate models with non-dynamic vegetation that show a surprising increase in leaf-area index—contradicting the hydrological trends observed over the region.

3.1.1 UKESM1-0-LL (r1i1p1f2) – Gradual decrease in tree fraction

The SSP5-8.5 scenario of this model shows a pronounced decline in tree fraction across eastern and central South America, decreasing from about 80% in the historical period (1850–1920) to around 45% by 2100 (Fig. 1). Spatial maps of the affected region indicate that trees are largely replaced by crops (Fig. 2). Since crop expansion is driven by prescribed land-use forcing rather than by climate dynamics, we do not classify this case as a climate-change response, and therefore it is not considered an SNS.

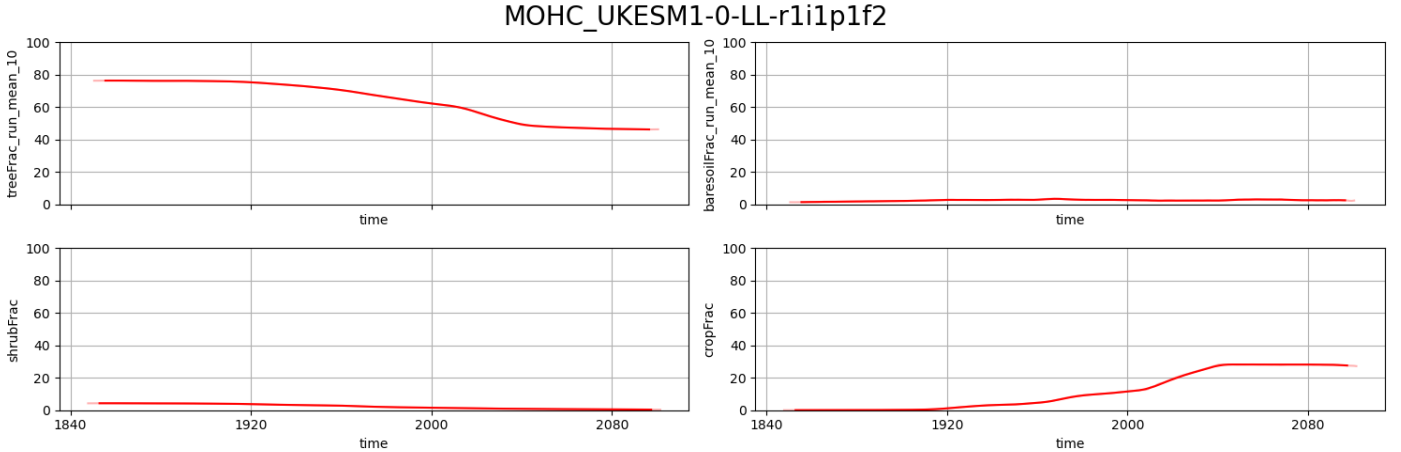


Figure 1: Time series of the case: gradual decreased tree fraction

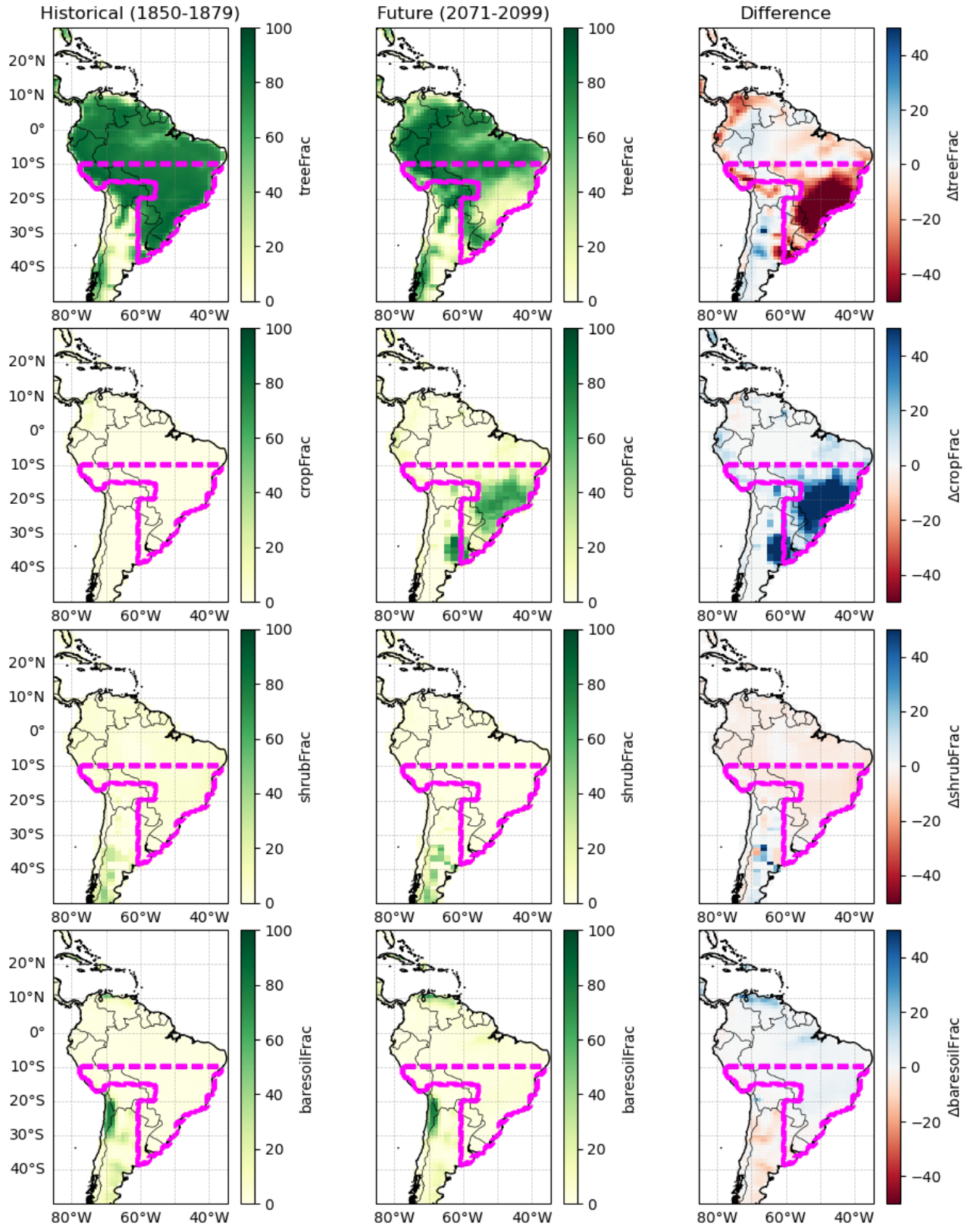


Figure 2: Spatial maps of the case: gradual decreased tree fraction

3.1.2 UKESM1-0-LL (r4i1p1f2) – Gradual decrease in leaf-area index and increase in bare-soil fraction

The SSP5-8.5 scenario of this model projects a gradual decline in leaf-area index over the Amazon basin (Figs. 3, 4). This decline is primarily driven by a reduction in tree fraction, which decreases from about 80% in the historical period to around 40% by 2300 (Figs. 3, 4). Consistently, the bare-soil area fraction increases from nearly zero to more than 40% (Figs. 3, 4). Other vegetation types—such as shrubs, crops, and pastures—show limited spatiotemporal variability over the region (Figs. 3, 4).

The reduction in tree fraction is also consistent with the decreased net primary production in both spatial maps and time series averaged over the region (Figs. 3, 4). The spatiotemporal evolution of precipitation over the region suggests that reduced rainfall over the Amazon may be driving the observed forest dieback (Figs. 3, 5). The region-averaged precipitation time series shows a sharp decline around 2030–2040, which may have been triggered by reduced moisture transport into the region by large-scale circulation patterns (Figs. 5, 6). This initial reduction in precipitation may have acted as a trigger for the decline in tree fraction (and leaf-area index), which in turn reduces region-mean evapotranspiration. The lower evapotranspiration then feeds back, further decreasing precipitation, which reinforces the decline in tree fraction. In this model, evapotranspiration accounts for about 65% of precipitation over the region, indicating a strong recycling mechanism.

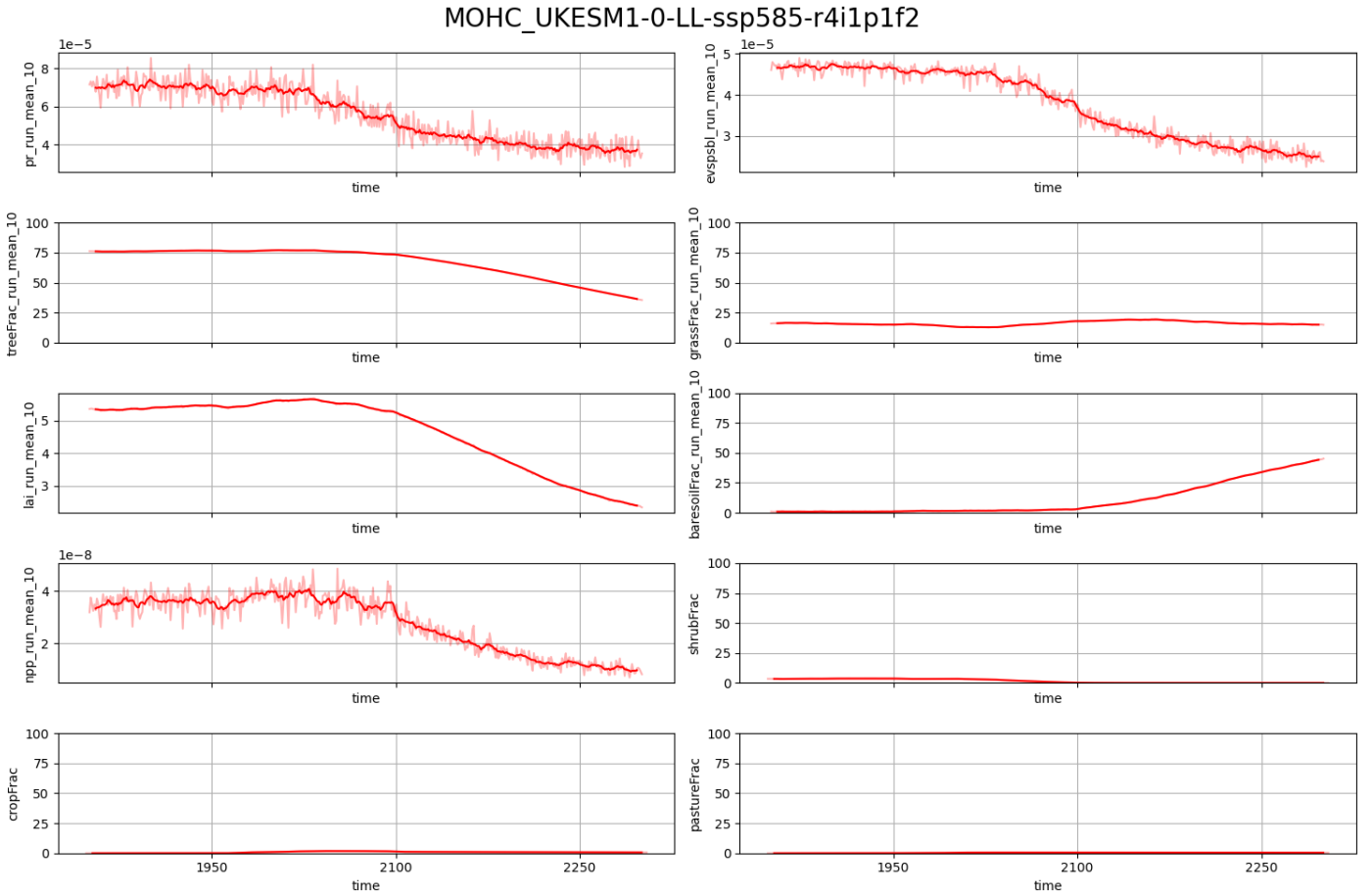


Figure 3: Time series of the case: gradual decrease in leaf-area index

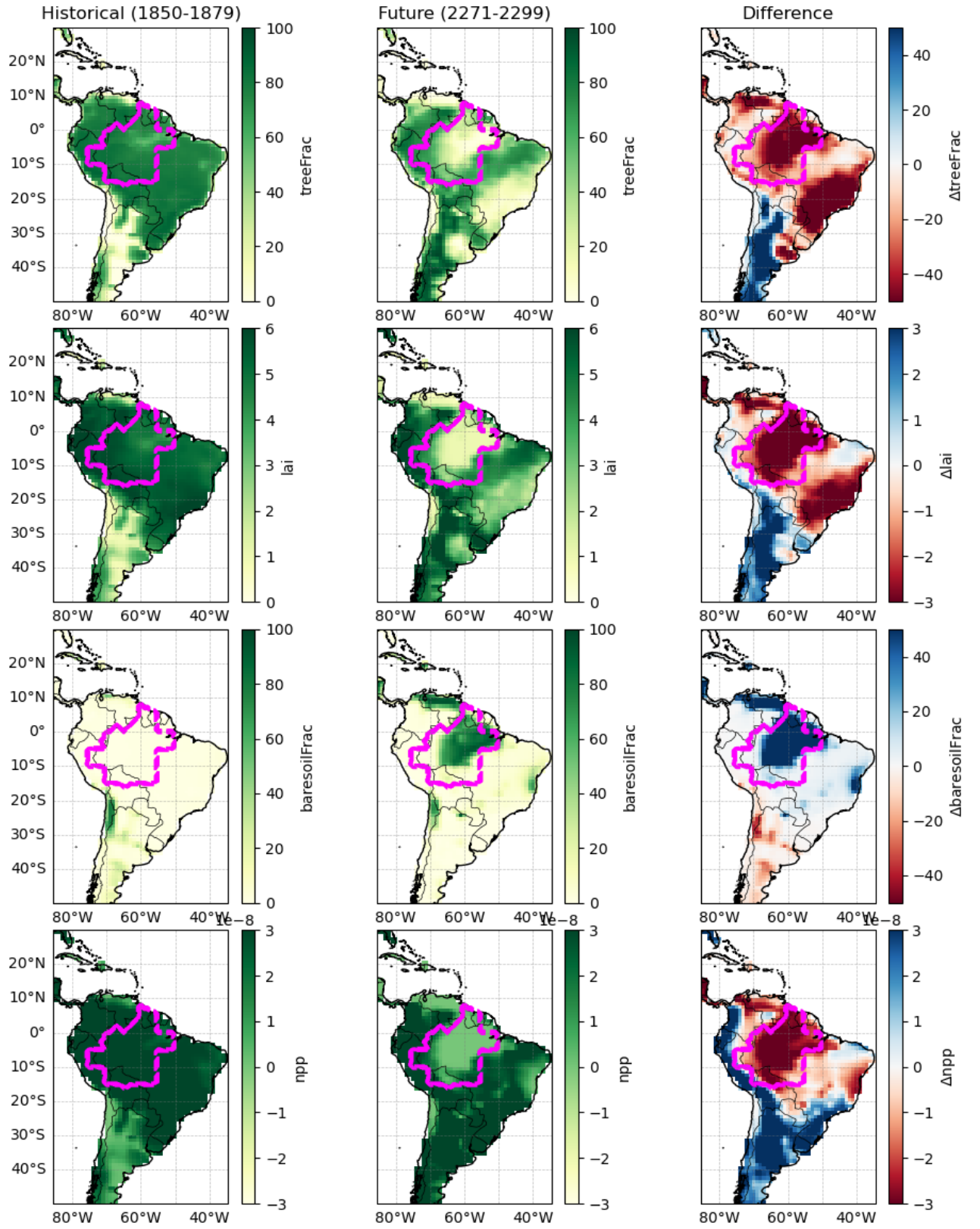


Figure 4: Spatial maps of the case: gradual decrease in leaf-area index

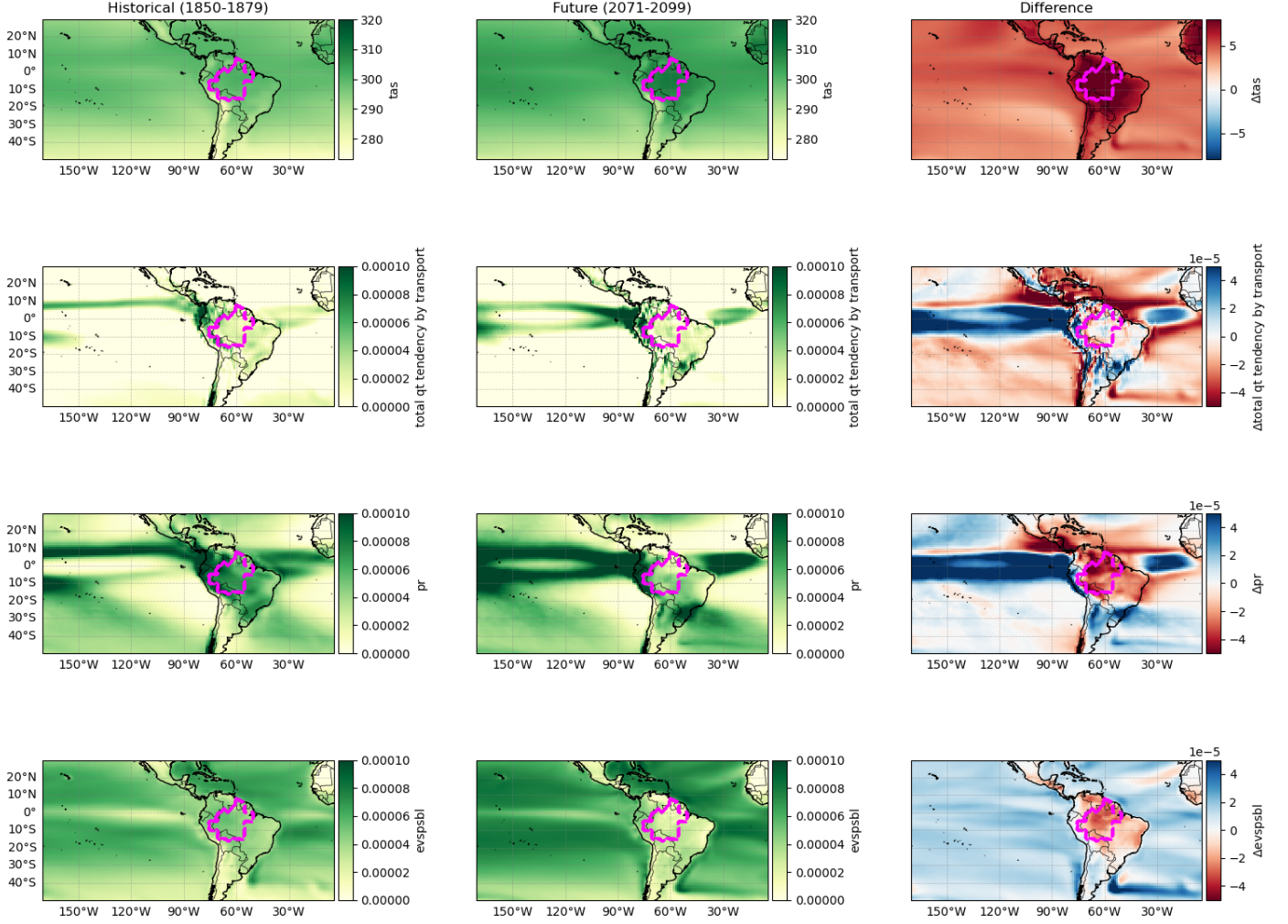


Figure 5: Spatial maps of the precipitation-related metrics for the case: gradual decrease in leaf-area index

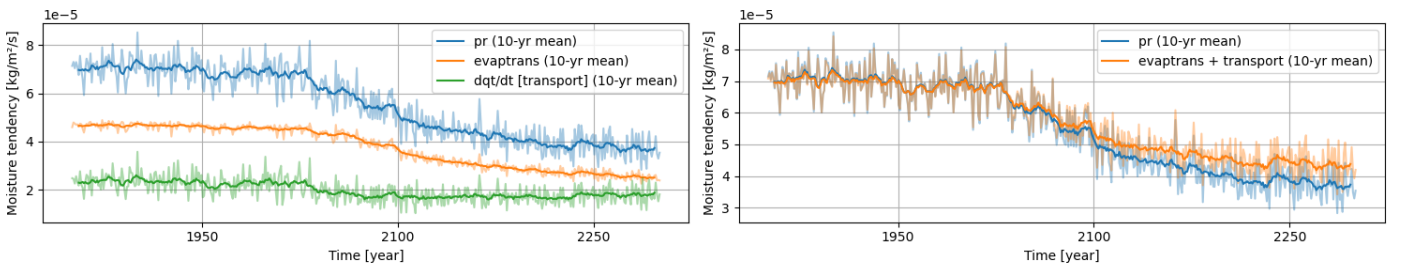


Figure 6: Time series of the precipitation-related metrics for the case: gradual decrease in leaf-area index

3.1.3 GFDL-ESM4 (r1i1p1f1) – Gradual decrease in tree fraction

The SSP3-7.0 scenario of this model shows a sudden, abrupt decrease in tree fraction, from around 50% to 30%, occurring around the year 2080 (Figs. 7, 8). A similar shift also occurs in the SSP5-8.5 scenario, albeit slightly earlier (around 2060). Crop and pasture fractions remain largely unchanged in this region, indicating that this change is not driven by land-use forcing. The decline in tree fraction appears to be triggered by a reduction in precipitation flux, which begins to decrease around 2020 and drives the subsequent drop in tree fraction (Figs. 7, 8). Consistently, evapotranspiration

decreases in response to reductions in tree fraction and leaf-area index (Figs. 7, 8).

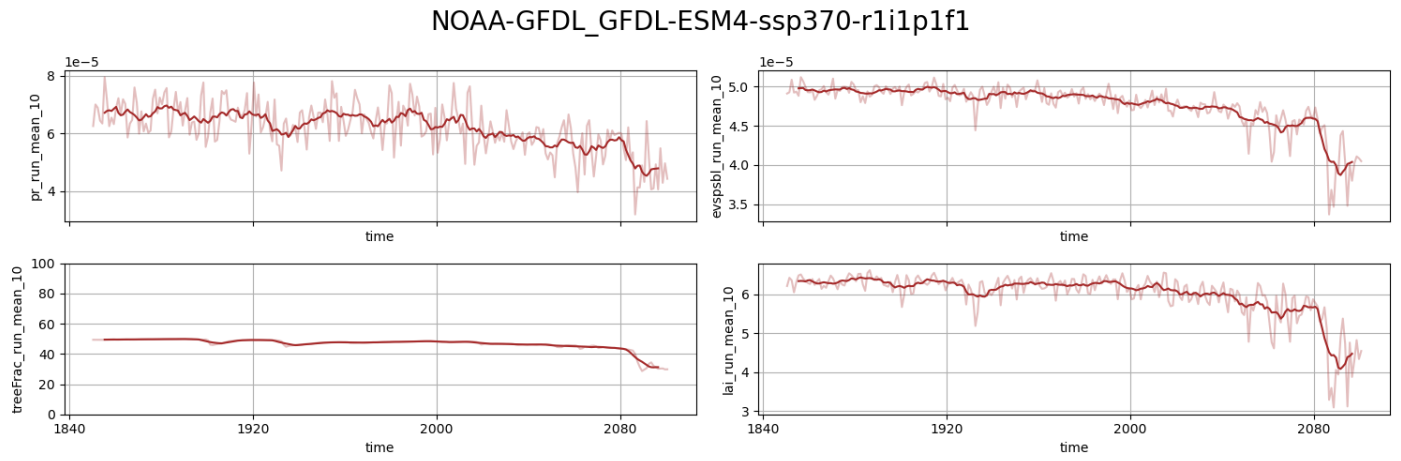


Figure 7: Time series of the case: gradual decrease in tree fraction

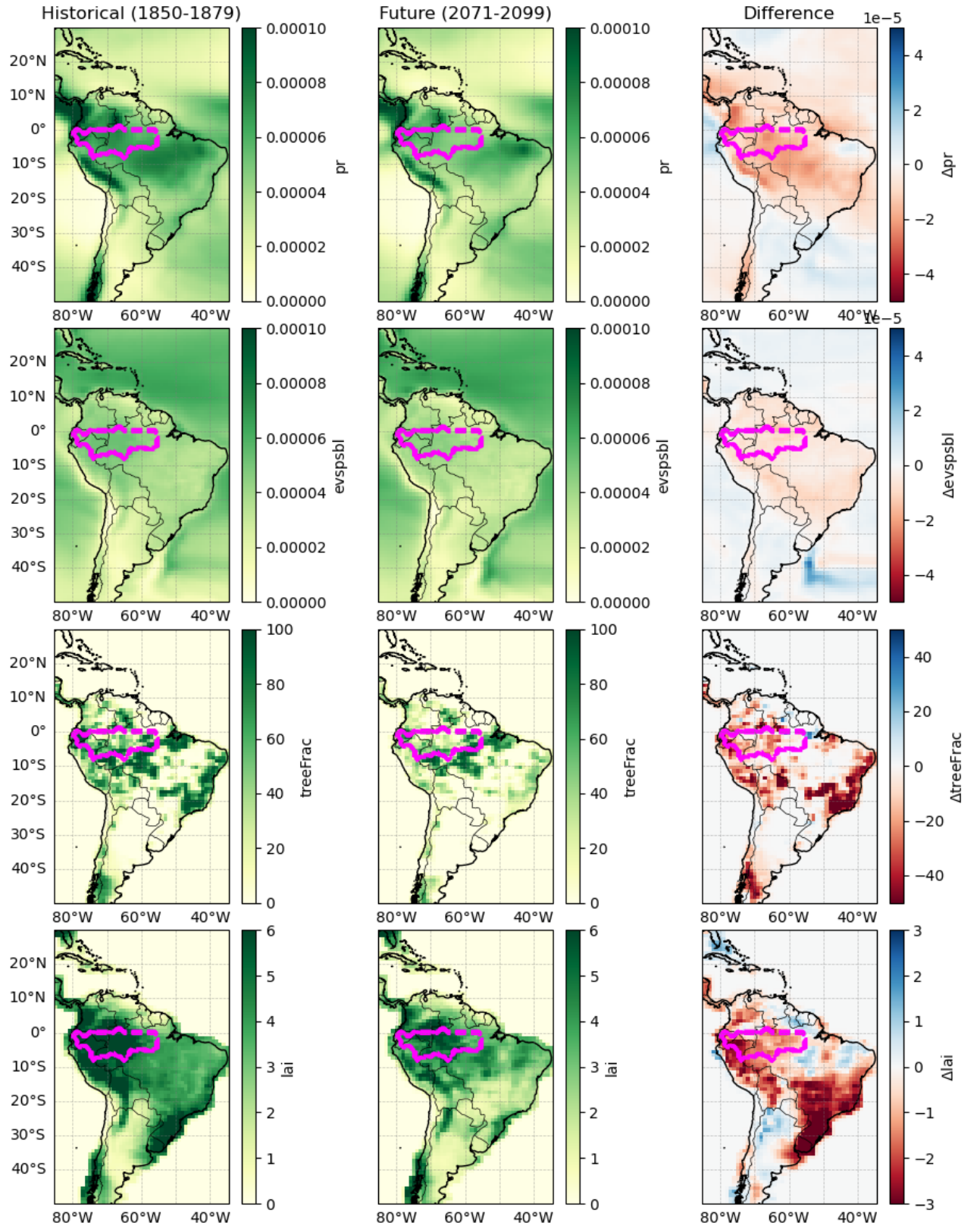


Figure 8: Spatial maps of the case: gradual decrease in tree fraction

3.1.4 IPSL-CM6A-LR (r1i1p1f1) – Gradual decrease in leaf-area index

The SSP5-8.5 scenario of this model shows a gradual decrease in leaf-area index, from values around 5 in the historical period to approximately 1.2 by 2300 (Fig. 9). This case clearly indicates that the reduction in precipitation occurs earlier than the decline in leaf-area index, suggesting that decreased precipitation is the primary driver of this shift. The reduction in evapotranspiration is more closely synchronized with the drop in leaf-area index. Since this model uses non-dynamic vegetation, vegetation types do not compete with one another, leaving the bare-soil fraction unchanged across time and space.

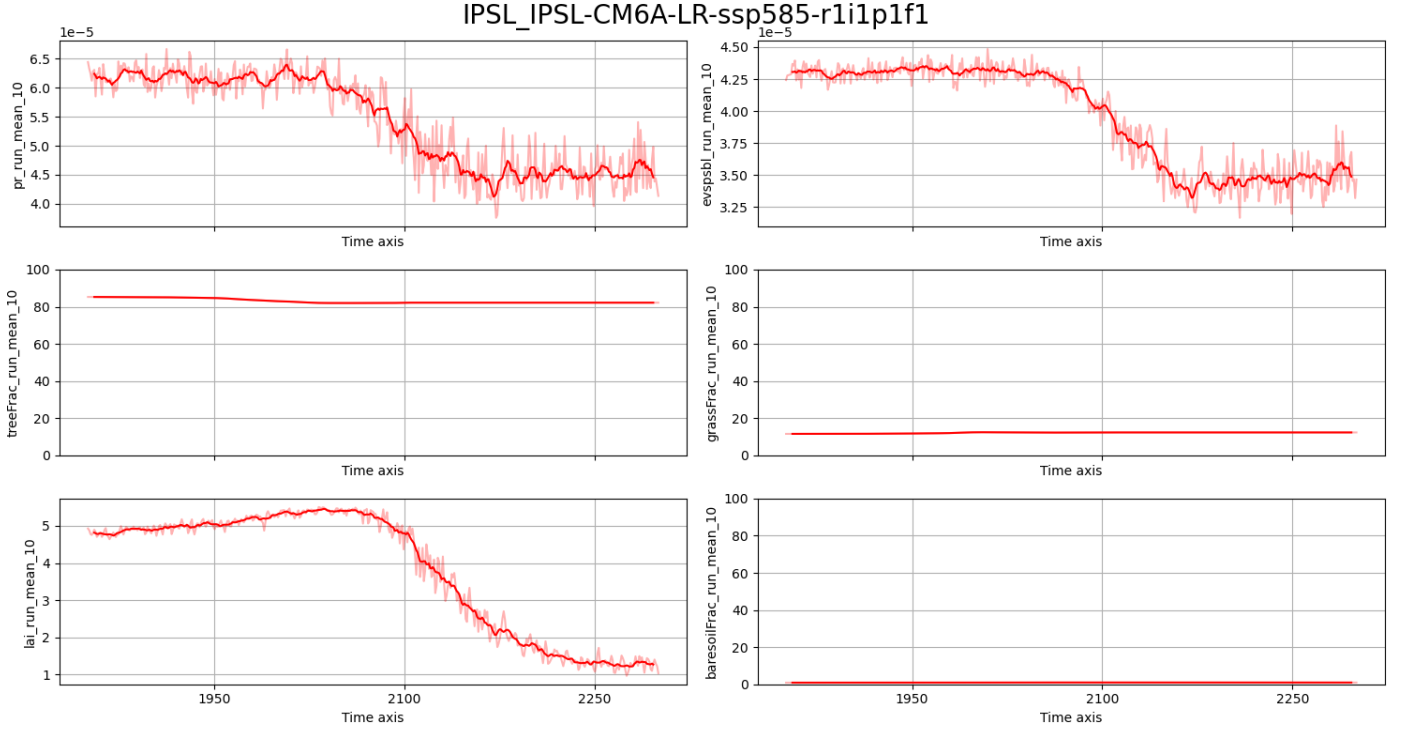


Figure 9: Time series of the case: gradual decrease in leaf-area index

3.1.5 Surprising increase of leaf-area index in three climate models

Three non-dynamic vegetation climate models—CESM2, CESM2-WACCM, and NorESM2-LM—project a strong increase in leaf-area index over the Amazon basin. This increase occurs despite decreased or unchanged precipitation over the region. In these cases, leaf-area index rises from around 4 in 1850 to approximately 10 by 2100, while precipitation and evapotranspiration either remain stable in time and space (NorESM2-LM) or even decrease (CESM2, CESM2-WACCM), which is shown by Figs. 10 and 11 for CESM2-WACCM. Both plots show that in the Amazonian basin for this model, leaf-area index and net primary production increase over time. If these outputs are not the result of errors, then the mechanism driving trees to become denser and develop thicker leaves—despite no change in precipitation or evapotranspiration—must be attributed to another factor, such as elevated CO_2 concentrations. In these models, higher CO_2 may create favorable conditions for plant growth, independently of changes in the water cycle.

NCAR_CESM2-WACCM-ssp585-r1i1p1f1

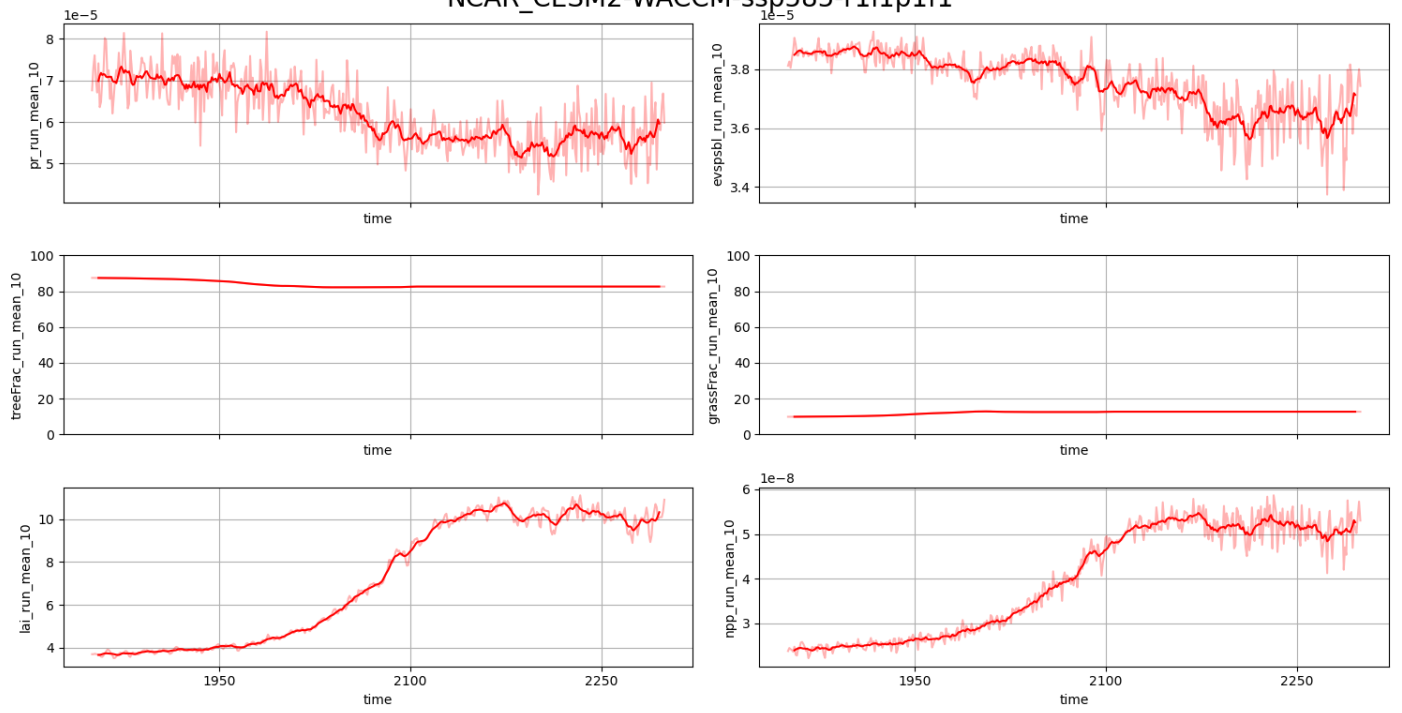


Figure 10: Time series of the case: gradual increase in leaf-area index

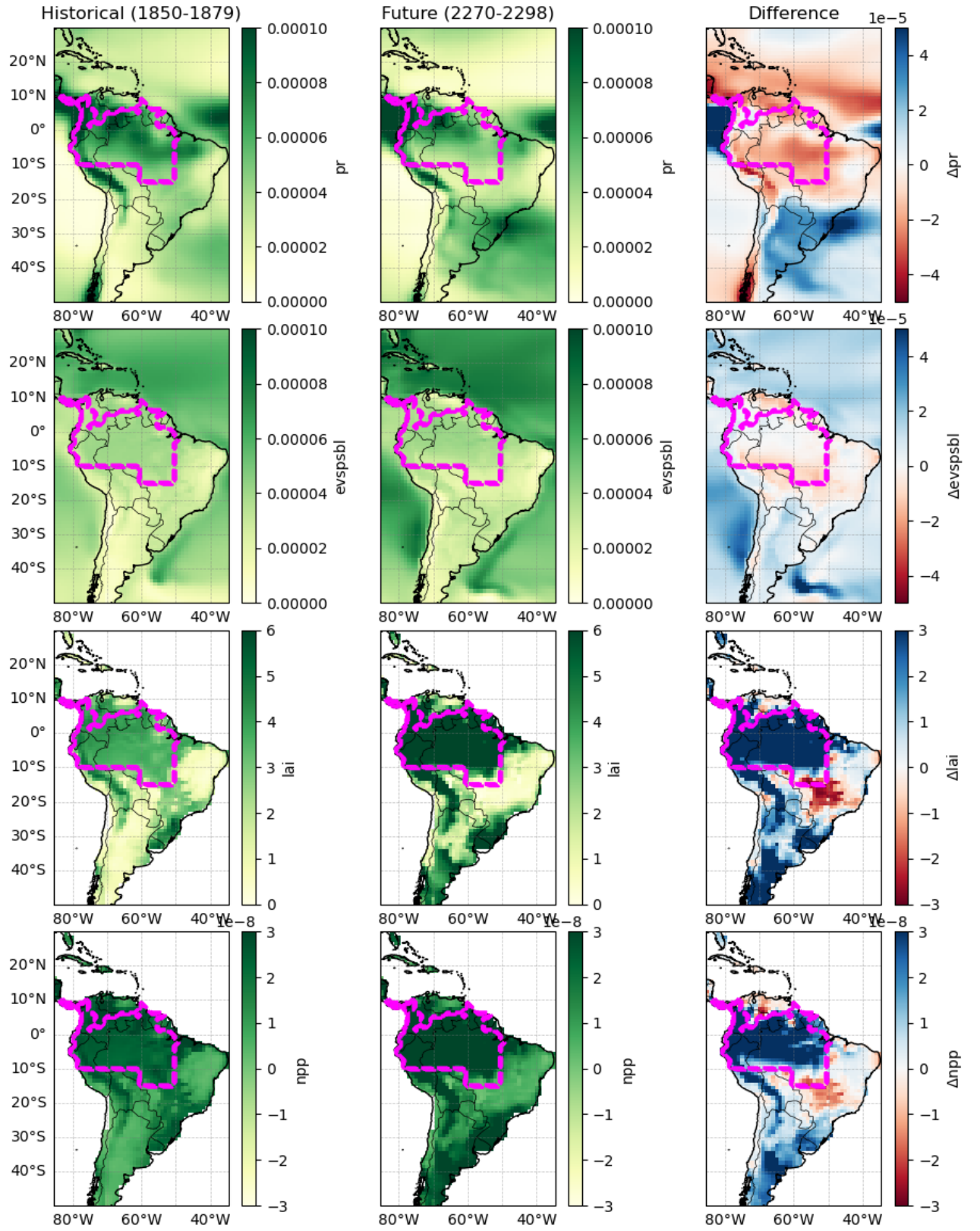


Figure 11: Spatial maps of the case: gradual increase in leaf-area index

3.2 African region

This section describes dramatic shifts over the African continent, ranging from a gradual decrease in bare-soil fraction in the eastern region and across the Sahel, to reductions in tree fraction and soil moisture in the African rainforest, known as the Congo basin.

3.2.1 UKESM1-0-LL (r4i1p1f2) – Gradual decrease in bare-soil fraction

The SSP5-8.5 scenario of this model projects a gradual decrease in bare-soil fraction over central and eastern Africa, from around 40% during 1850–2060 to approximately 10% by 2300 (Figs. 12, 13). This change is primarily driven by an increase in crop-area fraction of about 20% from 1850 to 2100, and partly by increases in tree (5–7%) and grass (10%) fractions from 2060 to 2300 (Figs. 12, 13). Although crop expansion is driven by the land-use change scenario in the southern part of the region, part of the decrease in bare-soil fraction is due to the increased tree and grass fractions in the northern part. Therefore, this case can still be considered an SNS (Figs. 12, 13).

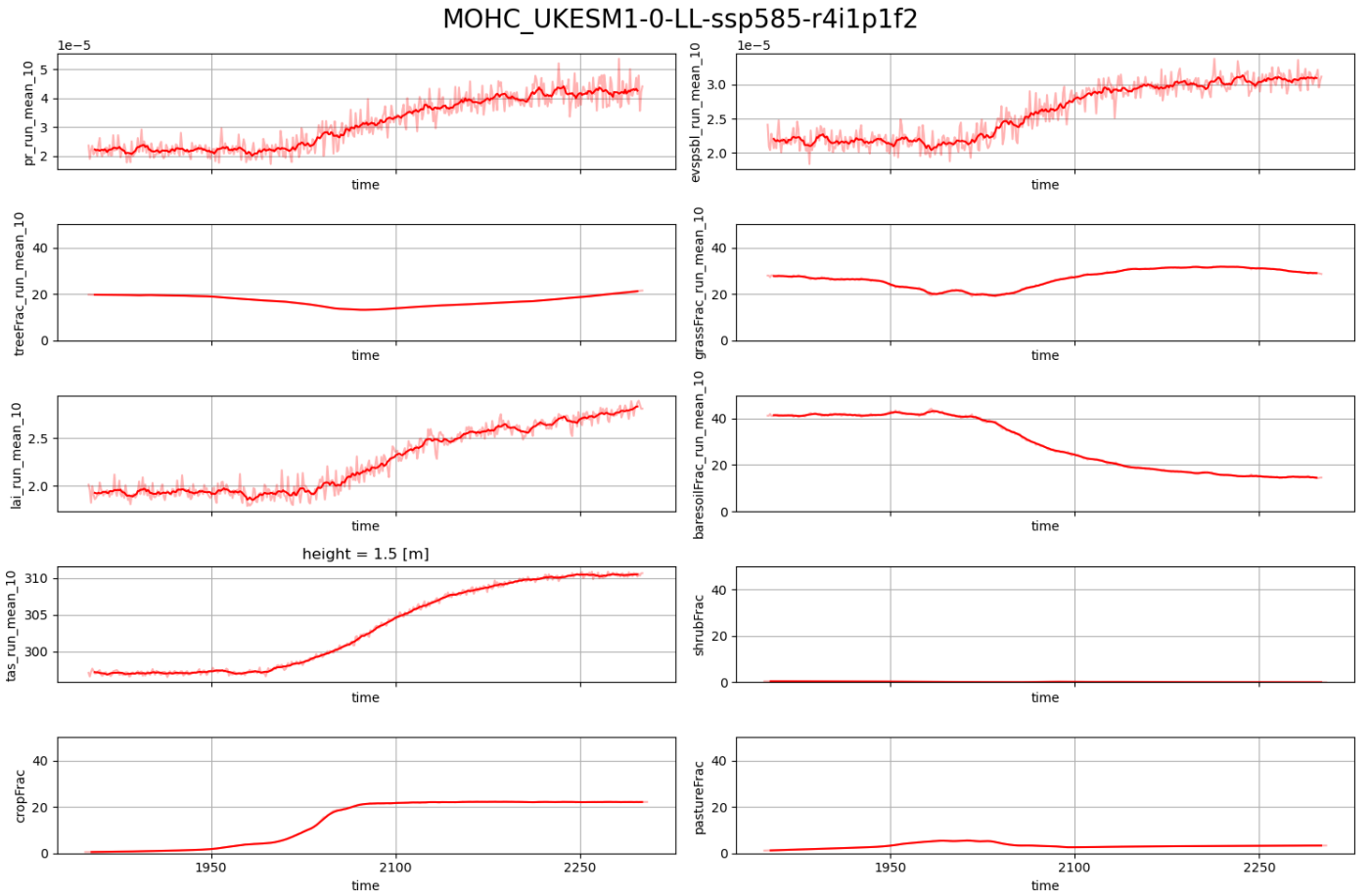


Figure 12: Time series of the case: gradual decrease in bare-soil fraction

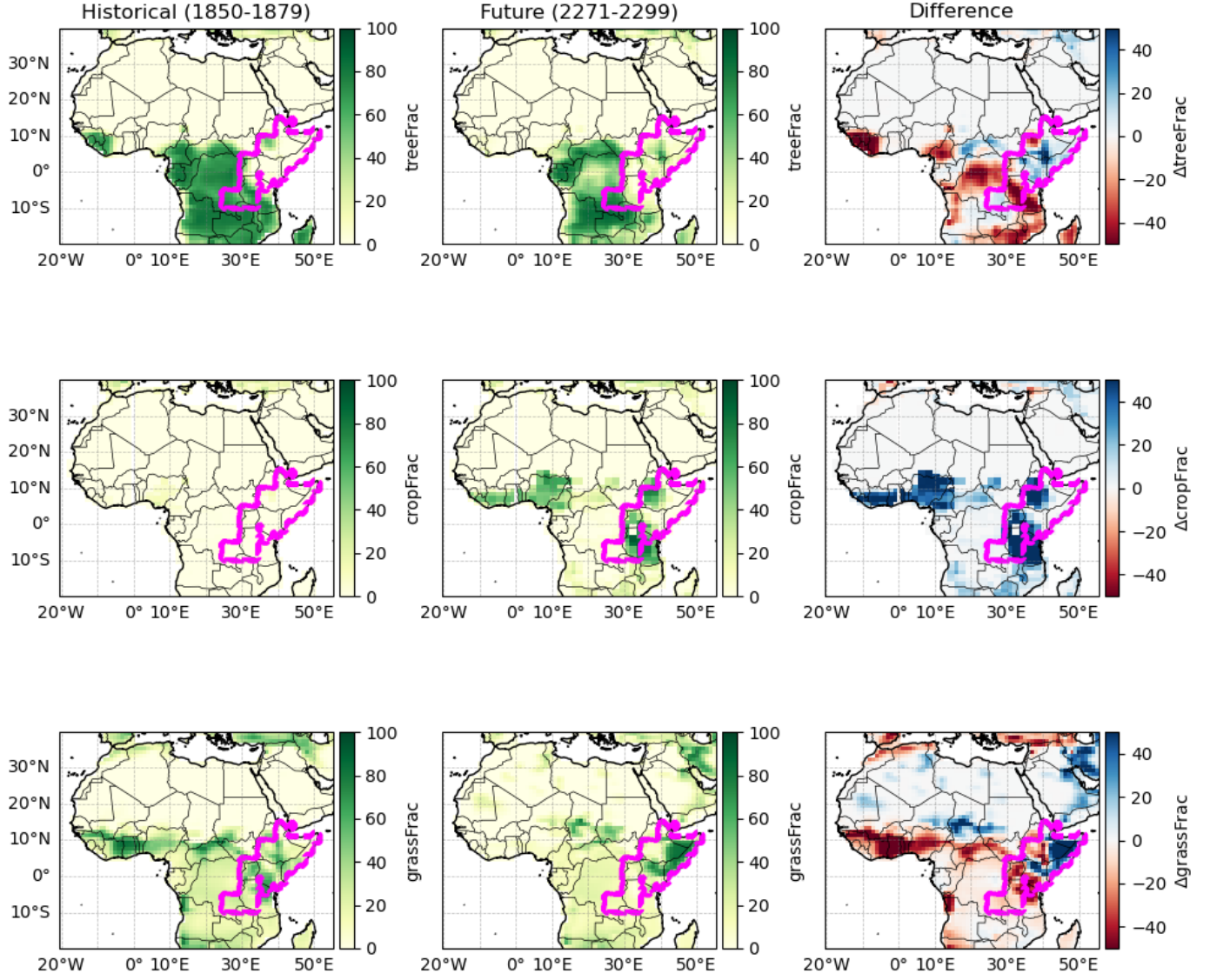


Figure 13: Spatial maps of the case: gradual decrease in bare-soil fraction

Over this region, precipitation begins to increase around 2015, which may contribute to the observed rise in tree fraction. This increase in precipitation can be primarily attributed to enhanced horizontal moisture transport into the region (Figs. 14, 15). The increasing tree fraction, in turn, drives higher evapotranspiration over the region, which feeds back to further increase precipitation (Figs. 14, 15).

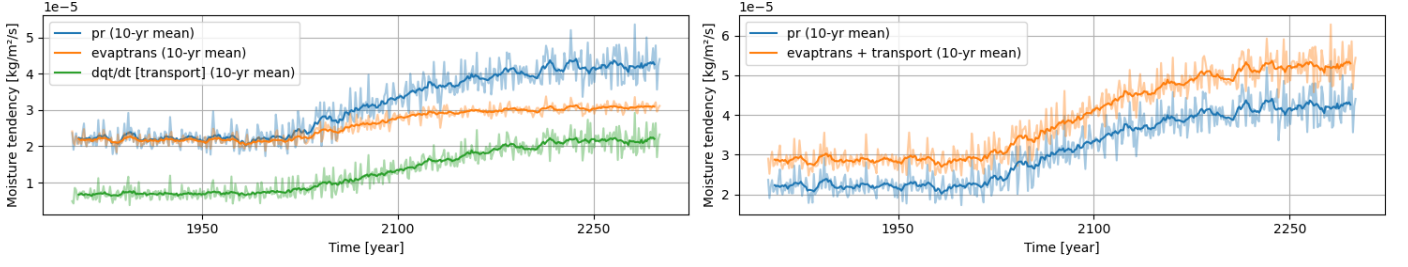


Figure 14: Time series of the precipitation-related metrics for the case: gradual decrease in bare-soil fraction

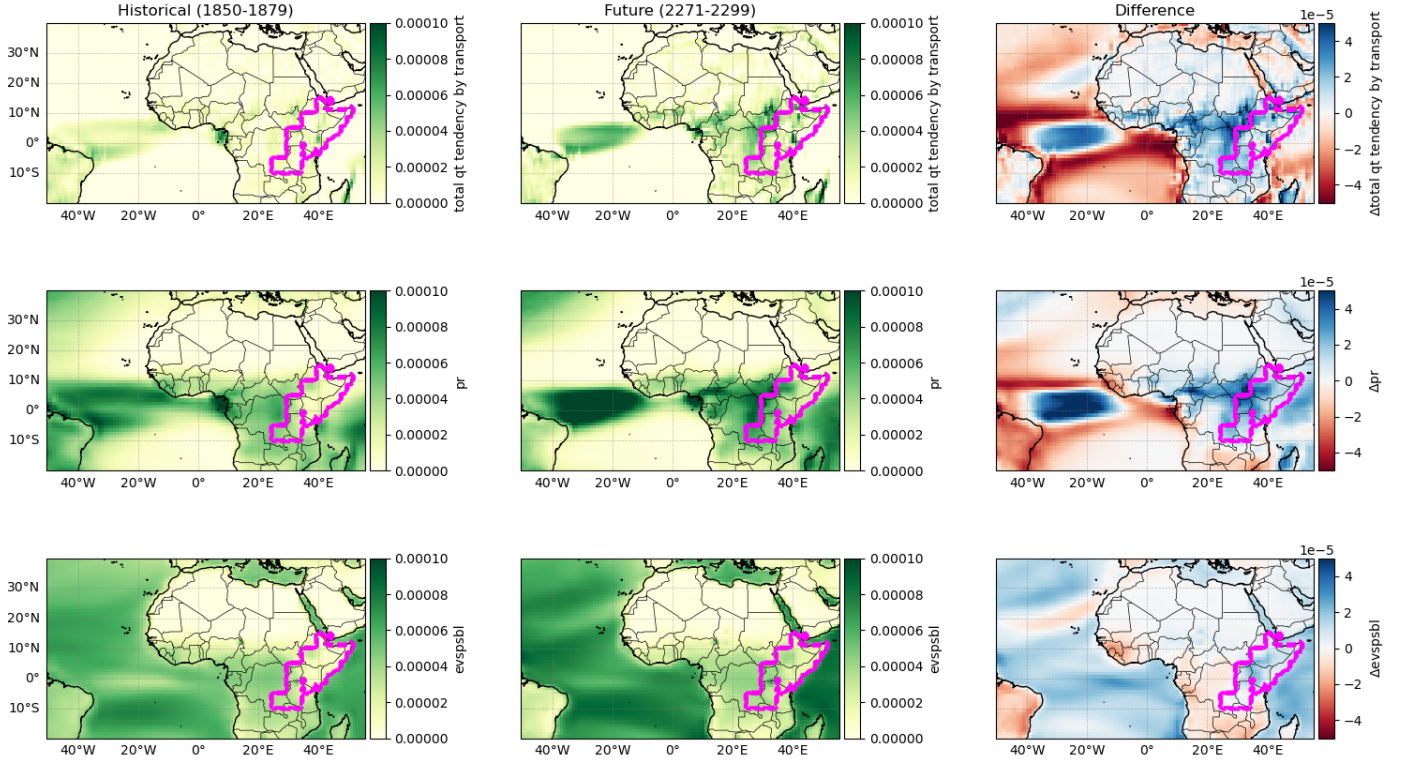


Figure 15: Spatial maps of the precipitation-related metrics for the case: gradual decrease in bare-soil fraction

3.2.2 UKESM1-0-LL (r16i1p1f2) – Gradual decrease in bare-soil fraction

The SSP3-7.0 scenario (as well as SSP5-8.5) of this model projects a gradual decrease in bare-soil fraction over the Sahel, from 90% to 60% by 2100 (Figs. 16, 17). This decrease is driven by an increase in crop fraction in the southwest of the region (about 20%) and an increase in grass fraction (10%) in the central part of the Sahel (Figs. 16, 17). While the increase in crop fraction results from land-use forcing, the increase in grass fraction is a natural response of the vegetation model, meaning that this case is still considered an SNS. Overall, precipitation slightly increases over the region, which may have created more favorable conditions for grass growth.

MOHC_UKESM1-0-LL-ssp370-r16i1p1f2

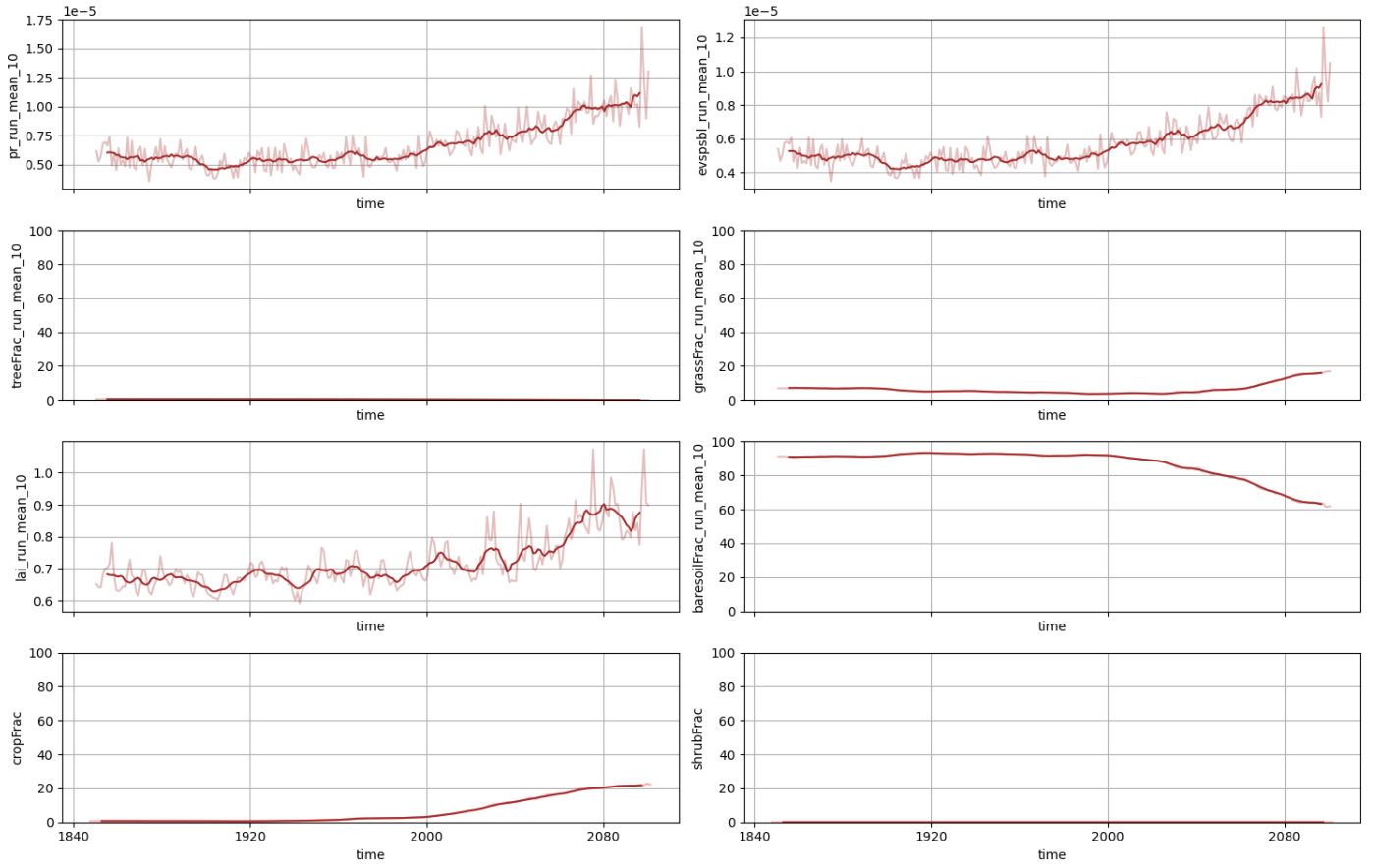


Figure 16: Time series of the case: gradual decrease in bare-soil fraction

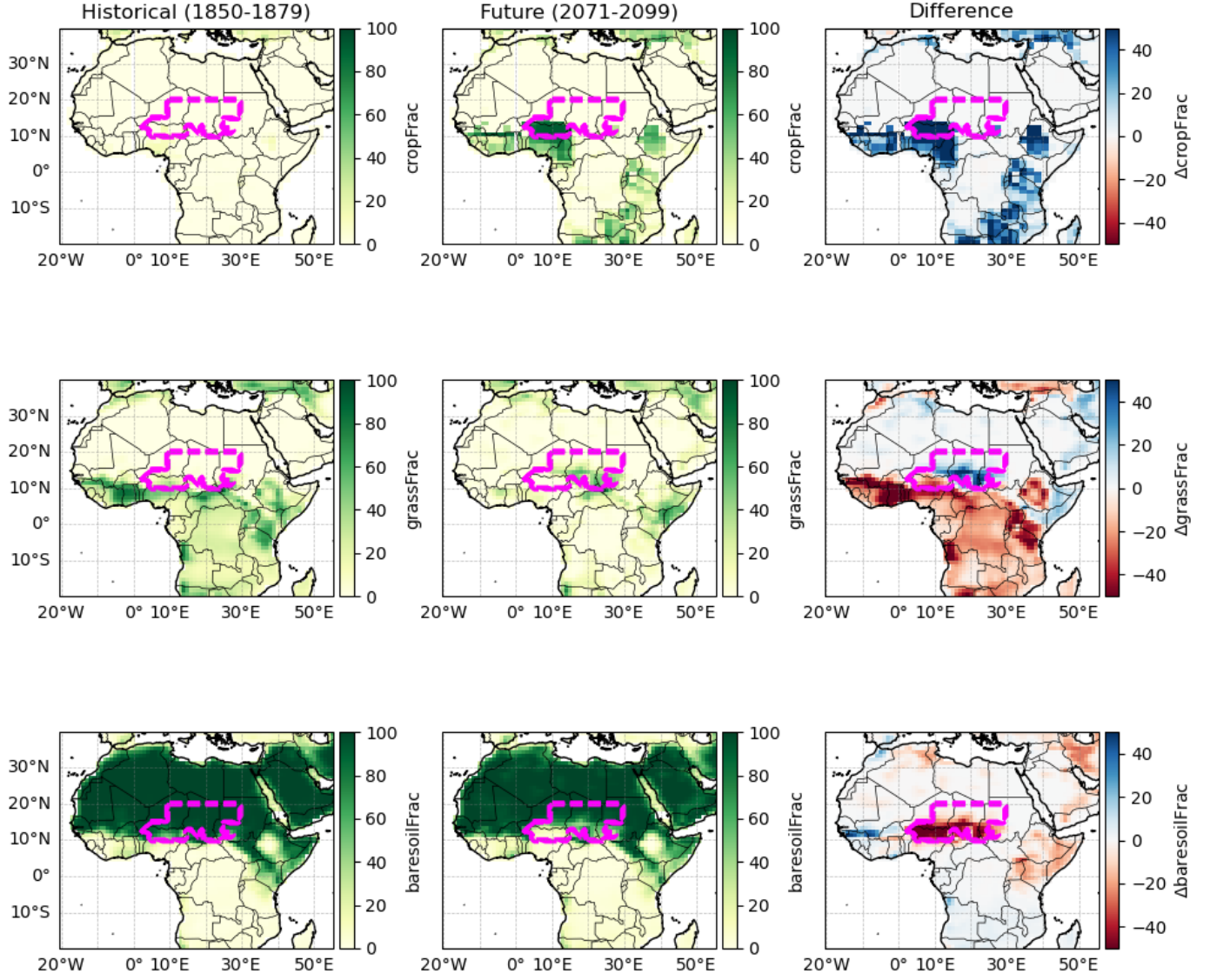


Figure 17: Spatial maps of the case: gradual decrease in bare-soil fraction

3.2.3 UKESM1-0-LL (r16i1p1f2) – Gradual decrease in tree fraction

The SSP3-7.0 scenario of this model shows a strong decline in tree coverage, from 80% to 20% by 2100 over the Congo basin (Figs. 18, 19). This change is primarily driven by an increase in pasture fraction as a result of the land-use change scenario. Therefore, we do not consider this case an SNS.

MOHC_UKESM1-0-LL-ssp370-r10i1p1f2

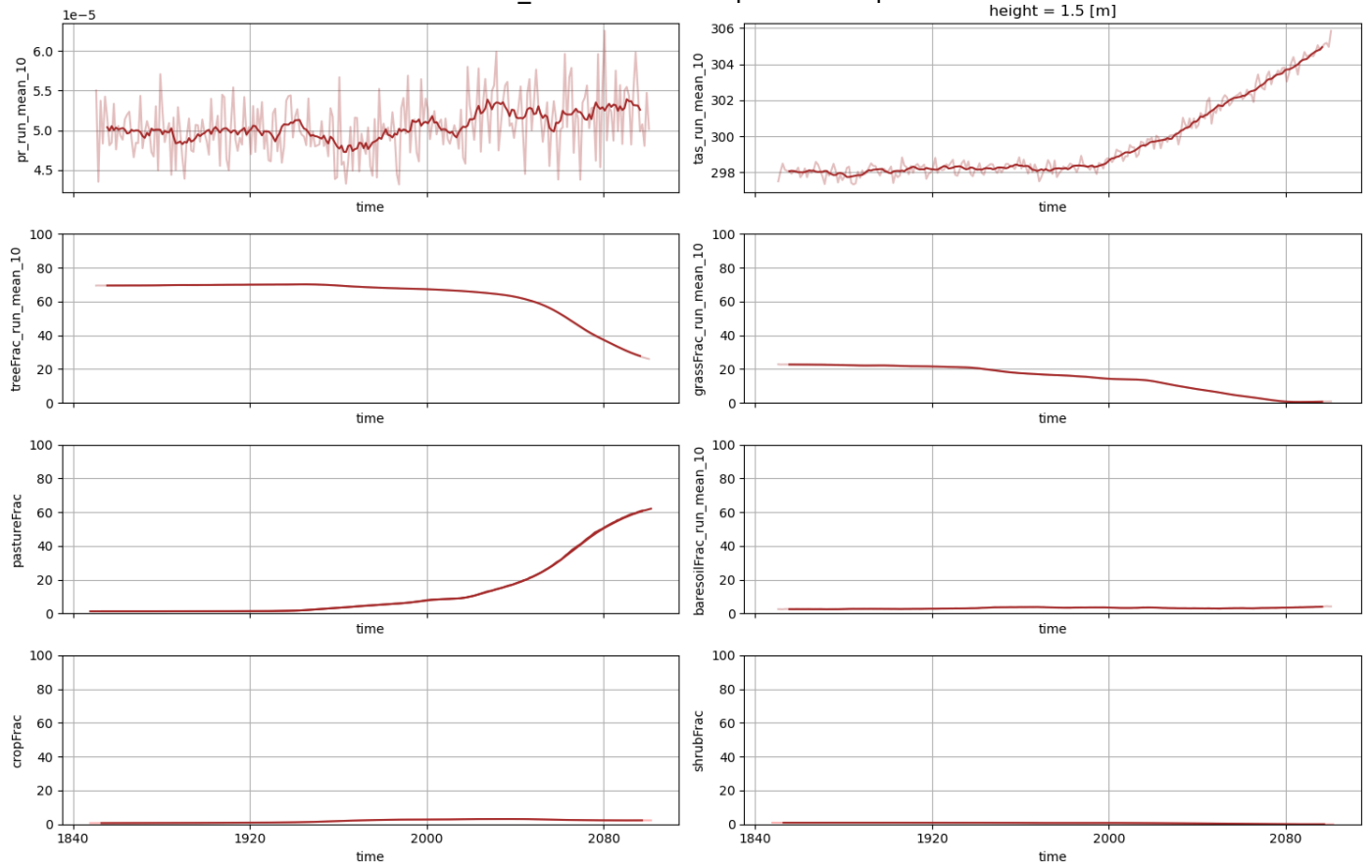


Figure 18: Time series of the case: gradual decrease in tree fraction

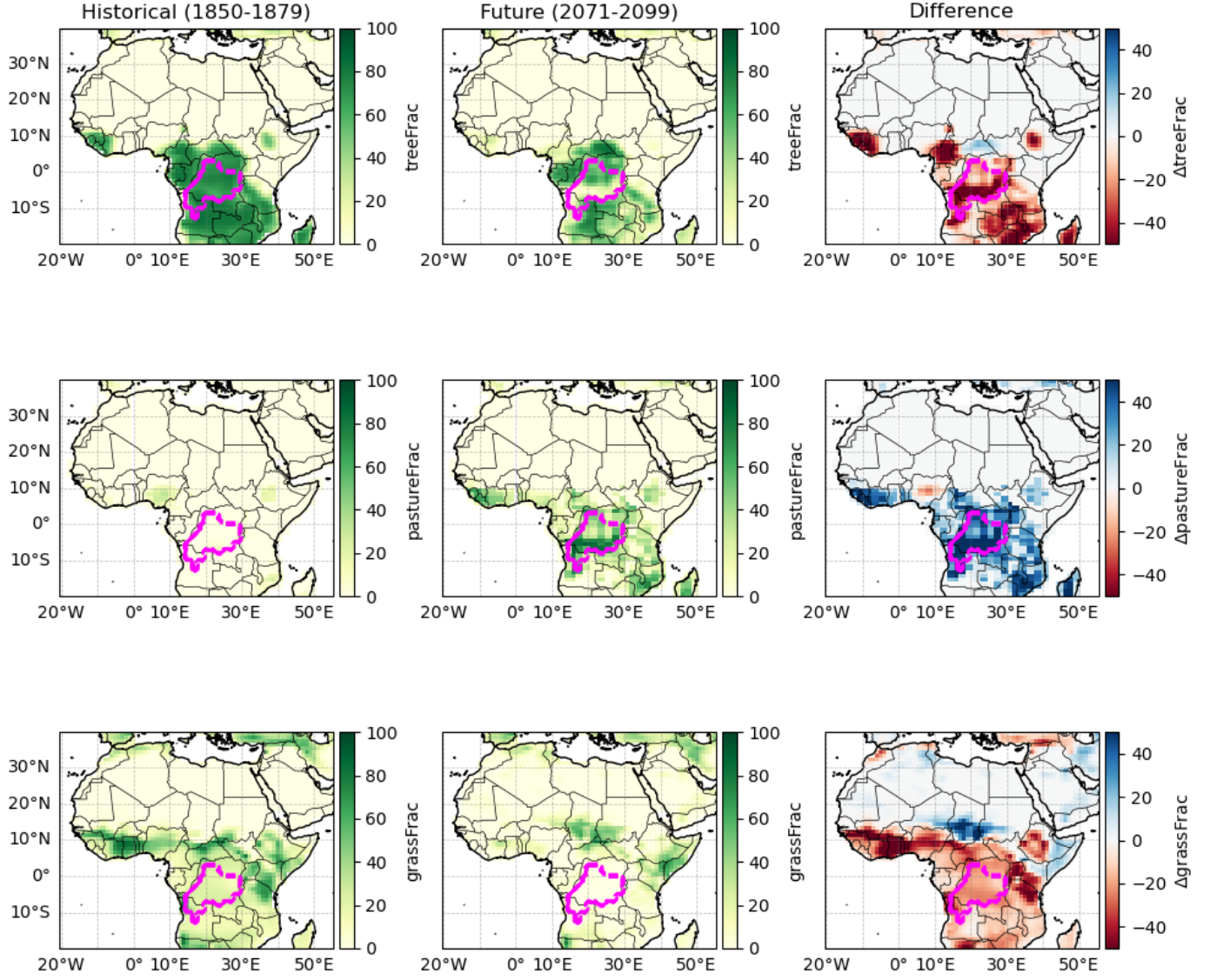


Figure 19: Spatial maps of the case: gradual decrease in tree fraction

3.2.4 Four non-dynamic vegetation models project an increased leaf-area index

The SSP5-8.5 scenario of the models CanESM5-CanOE, CanESM5-1, CanESM5, and NorESM2-LM projects a gradual increase in leaf-area index in the Congo basin, where the increase in precipitation may be the primary driver of this transition (see the example plots for CanESM5-CanOE in Figs. 20, 21). Consistently, evapotranspiration also increases, likely as a result of the higher leaf-area index.

CCCma_CanESM5-CanOE-ssp585-r1i1p2f1

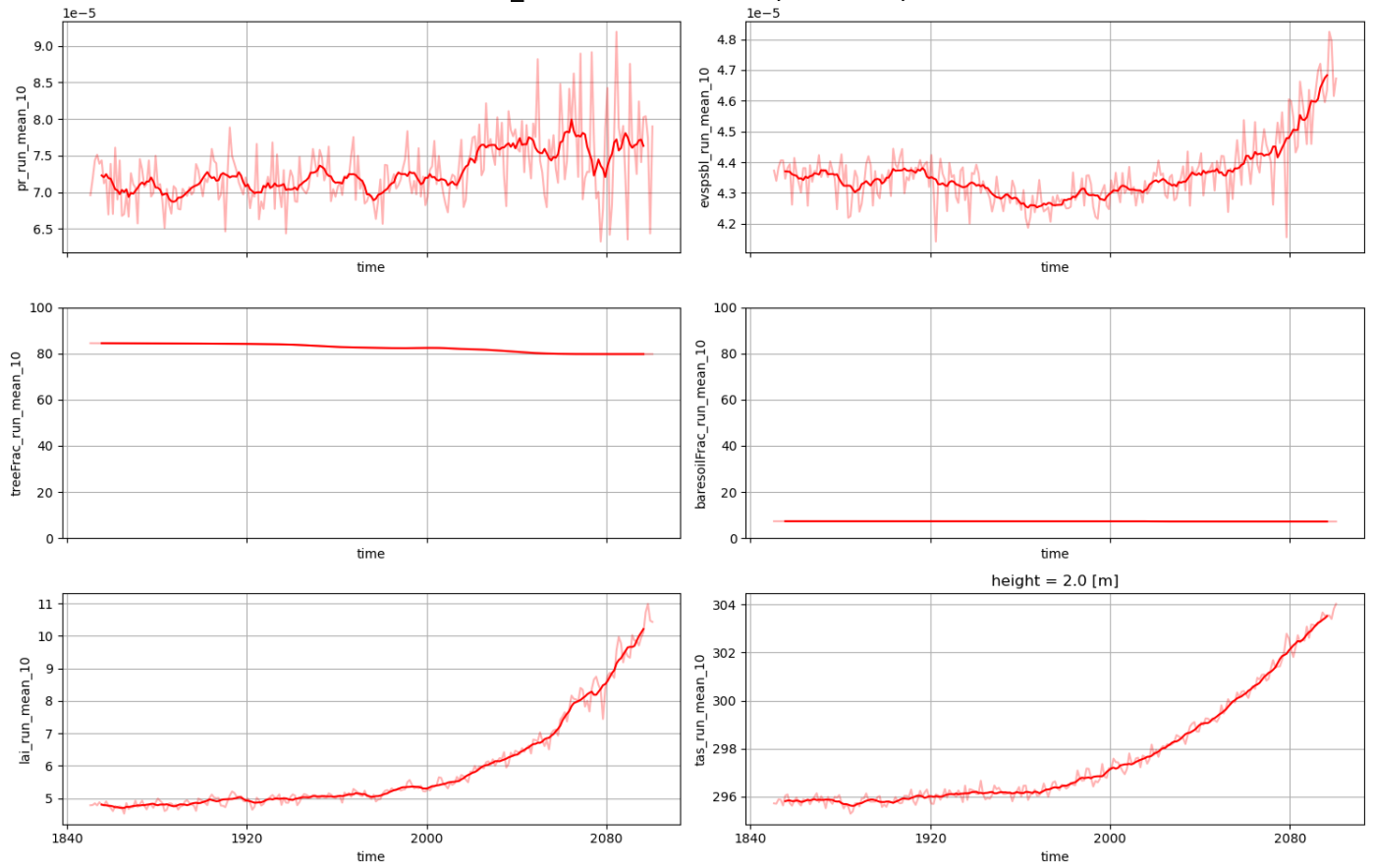


Figure 20: Time series of the case: gradual increase in leaf-area index

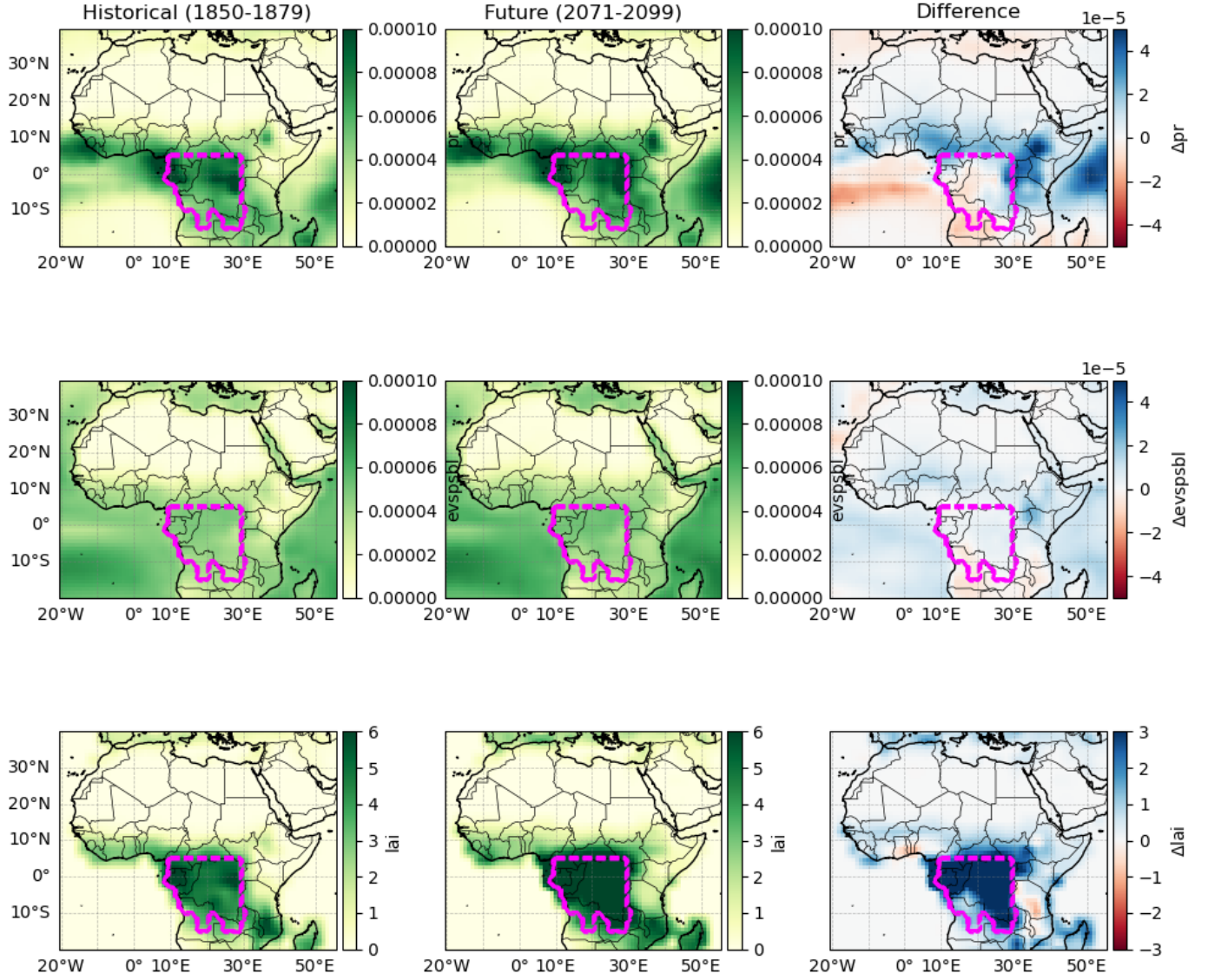


Figure 21: Spatial maps of the case: gradual increase in leaf-area index

3.2.5 MRI-ESM2-0 (r2i1p1f1) – Abrupt decrease in soil moisture

The SSP5-8.5 scenario of this model shows an abrupt decrease in soil moisture over the Congo basin, from 700 to 500 kg m⁻² around 2040 (Fig. 22). Despite considerable yearly variability in precipitation and evapotranspiration over the region, both exhibit reductions around the same period (Fig. 22). Spatial maps comparing the first and last thirty years of the model indicate that the decrease in soil moisture occurs exactly in the same areas where precipitation and evapotranspiration declined, suggesting that the reduction in precipitation may be driving this shift (Fig. 23). Another possibility is that increased temperature over the same region is the primary driver of the soil moisture decline, with the decreases in evapotranspiration and precipitation occurring as a response to reduced soil moisture.

MRI_MRI-ESM2-0-ssp585-r2i1p1f1

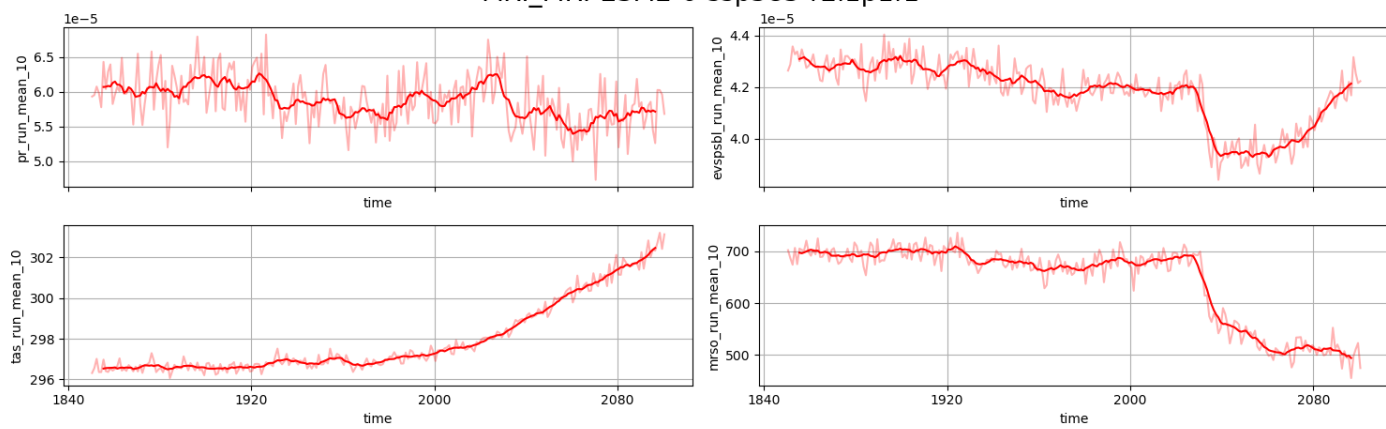


Figure 22: Time series of the case: abrupt decrease in soil moisture

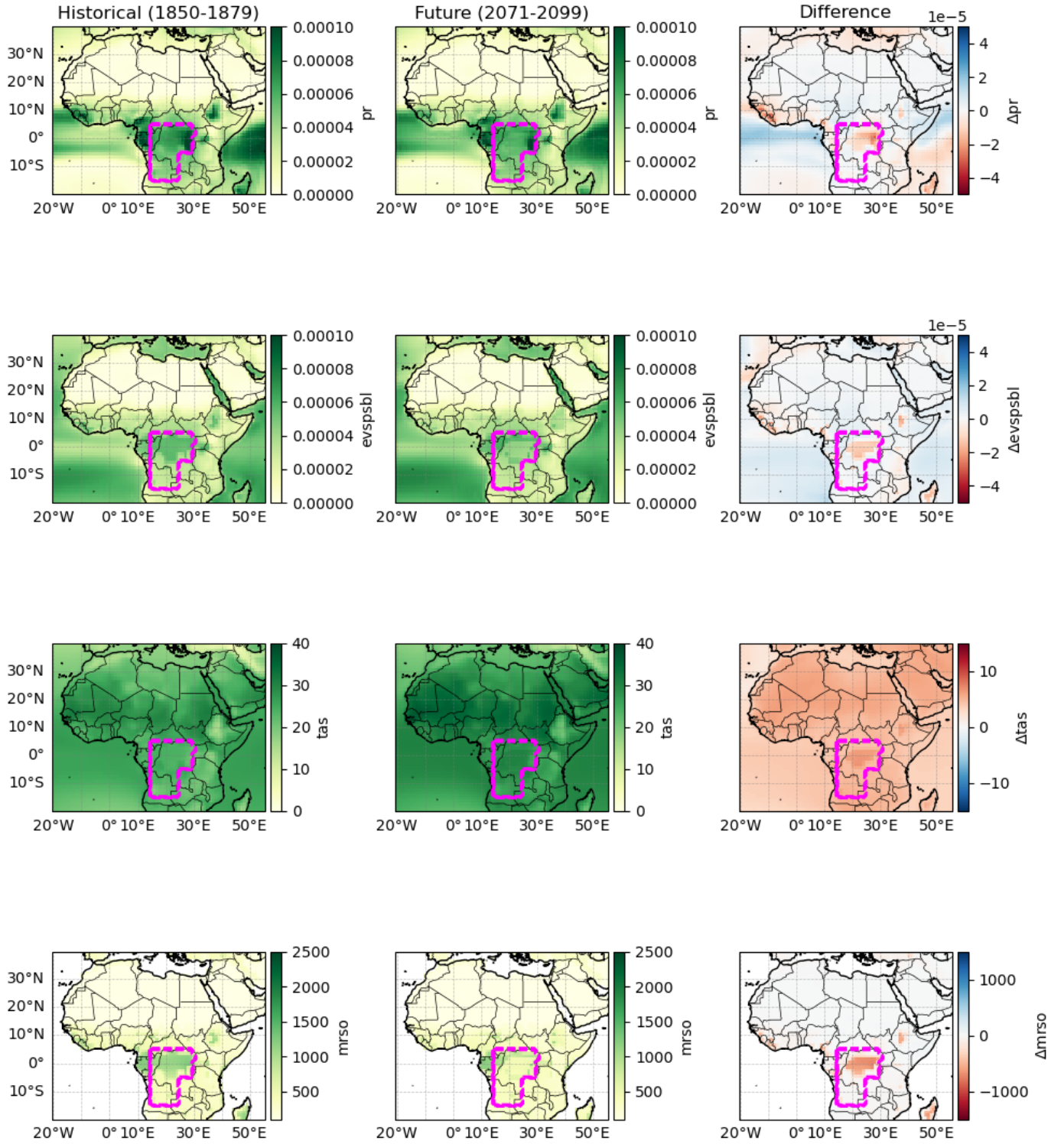


Figure 23: Spatial maps of the case: abrupt decrease in soil moisture

3.2.6 CNRM-ESM2-1 (r3i1p1f2) – Gradual decrease in soil moisture

The SSP3-7.0 scenario of this model shows a gradual decrease in soil moisture, from 1400 kg m^{-2} during 1850–2040 to 600 kg m^{-2} by 2080 (Figs. 24, 25). Synchronously, tree fraction decreases by 50% by 2100, while grass fraction increases, indicating a transition in vegetation type over the region (Figs. 24, 25). Although this analysis does not determine causality, decreased soil moisture may drive the transition from trees to grass, and conversely, the replacement of trees by grass can amplify soil moisture loss, since soil covered with grass is more susceptible to evaporation than soil covered with trees.

CNRM-CERFACS_CNRM-ESM2-1-ssp370-r3i1p1f2

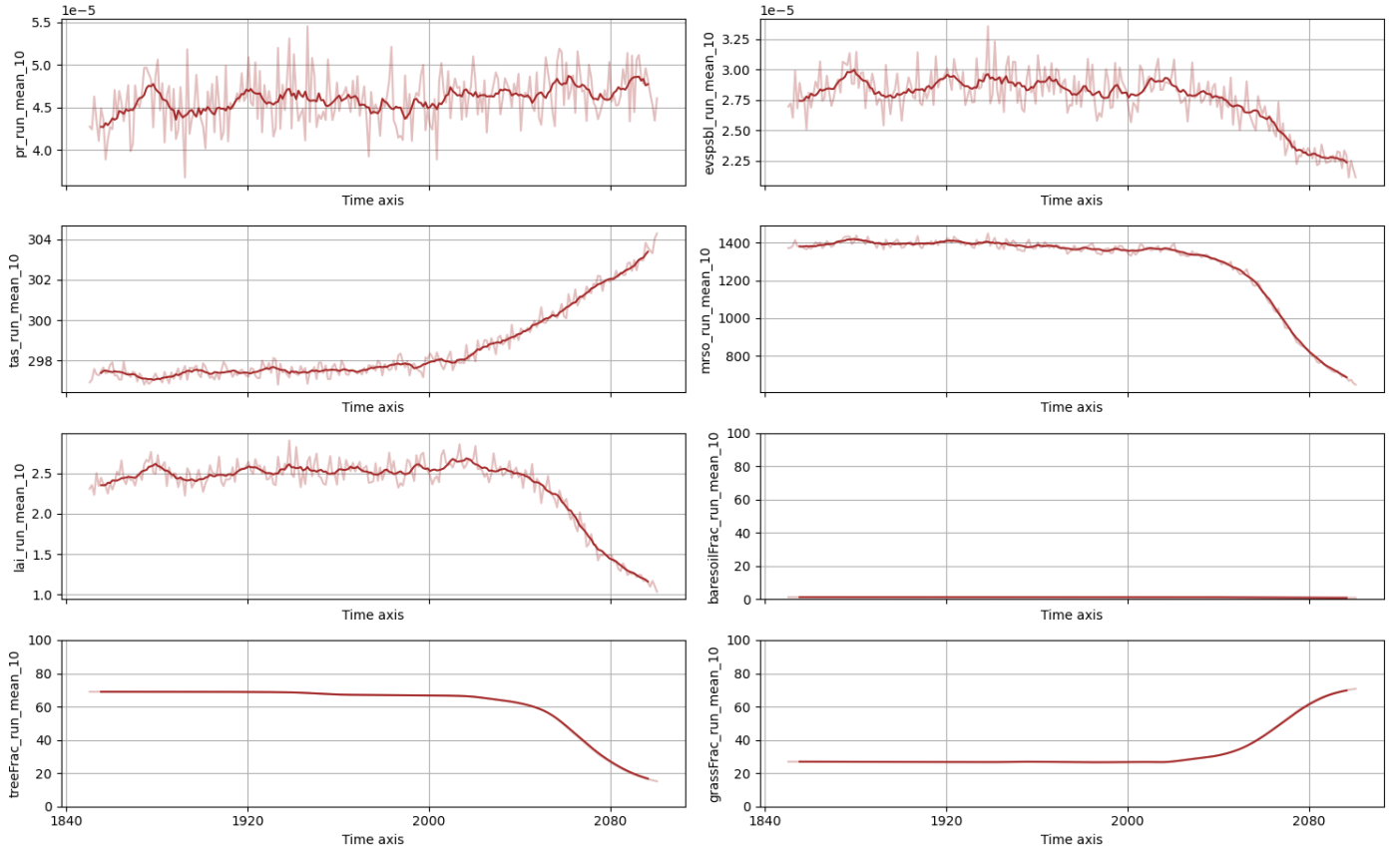


Figure 24: Time series of the case: gradual decrease in soil moisture

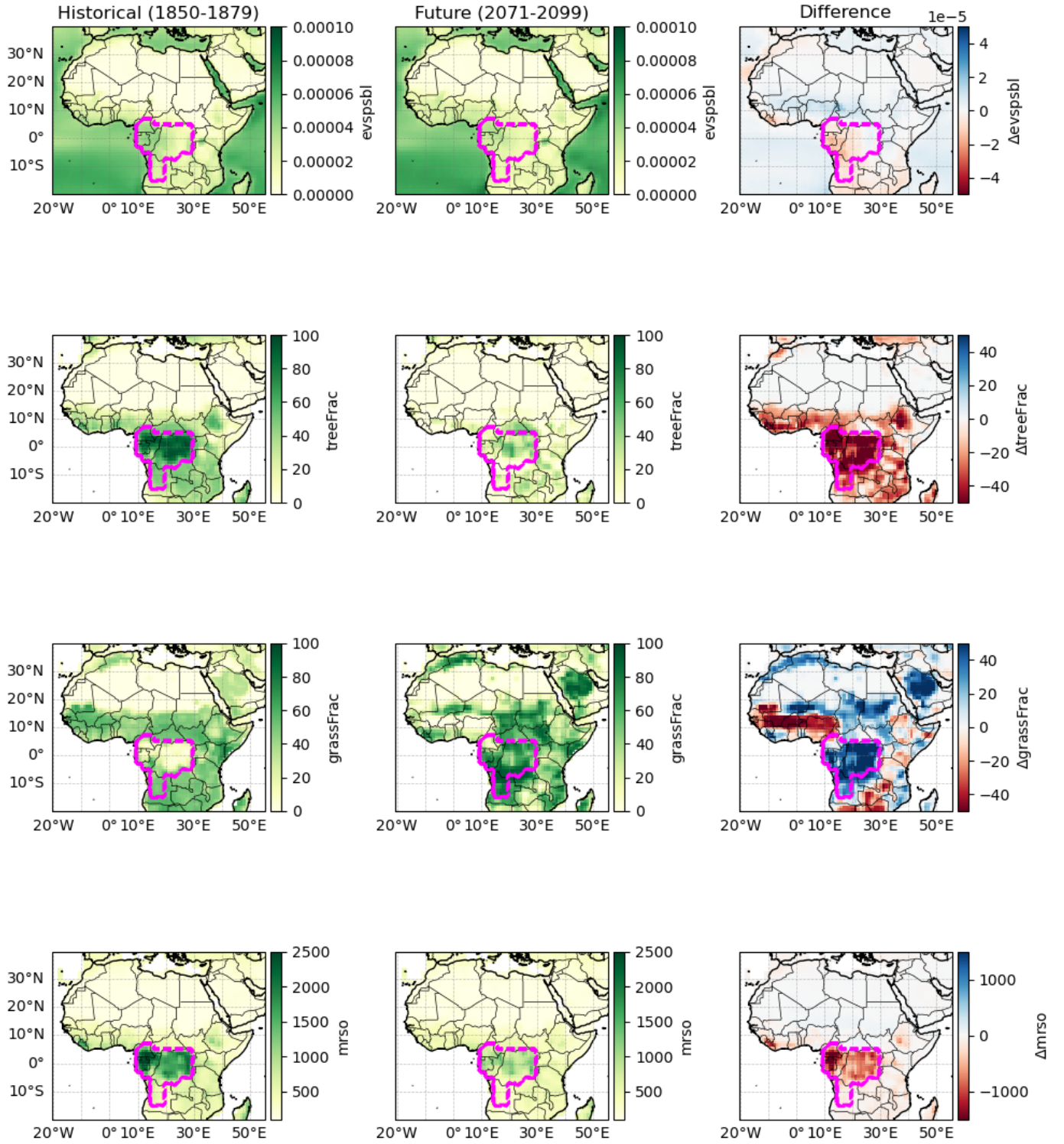


Figure 25: Spatial maps of the case: gradual decrease in soil moisture

3.3 Eastern Asia and North America

In this section, we present eight cases over China and two cases over North America in which vegetation-type area coverage exhibits SNSs.

3.3.1 UKESM1-0-LL (r1i1p1f2) – Gradual decrease in bare-soil fraction

The SSP5-8.5 scenario of this model projects a gradual decrease in bare-soil fraction over the southeastern part of China, from 80% during 1850–1920 to around 30% by 2100 (Figs. 26, 27). Both the time series and spatial maps indicate that this decrease is primarily driven by an increase in grass fraction over the region (Figs. 26, 27). The fourth member (r4i1p1f2), which extends to 2300, also projects the same trend in bare-soil and grass fractions. Additionally, there is a small increase in precipitation and evapotranspiration over the region, which may create more favorable conditions for grass growth. Tree fraction also shows a very slight increase, reaching about 6% by 2300 in the fourth member.

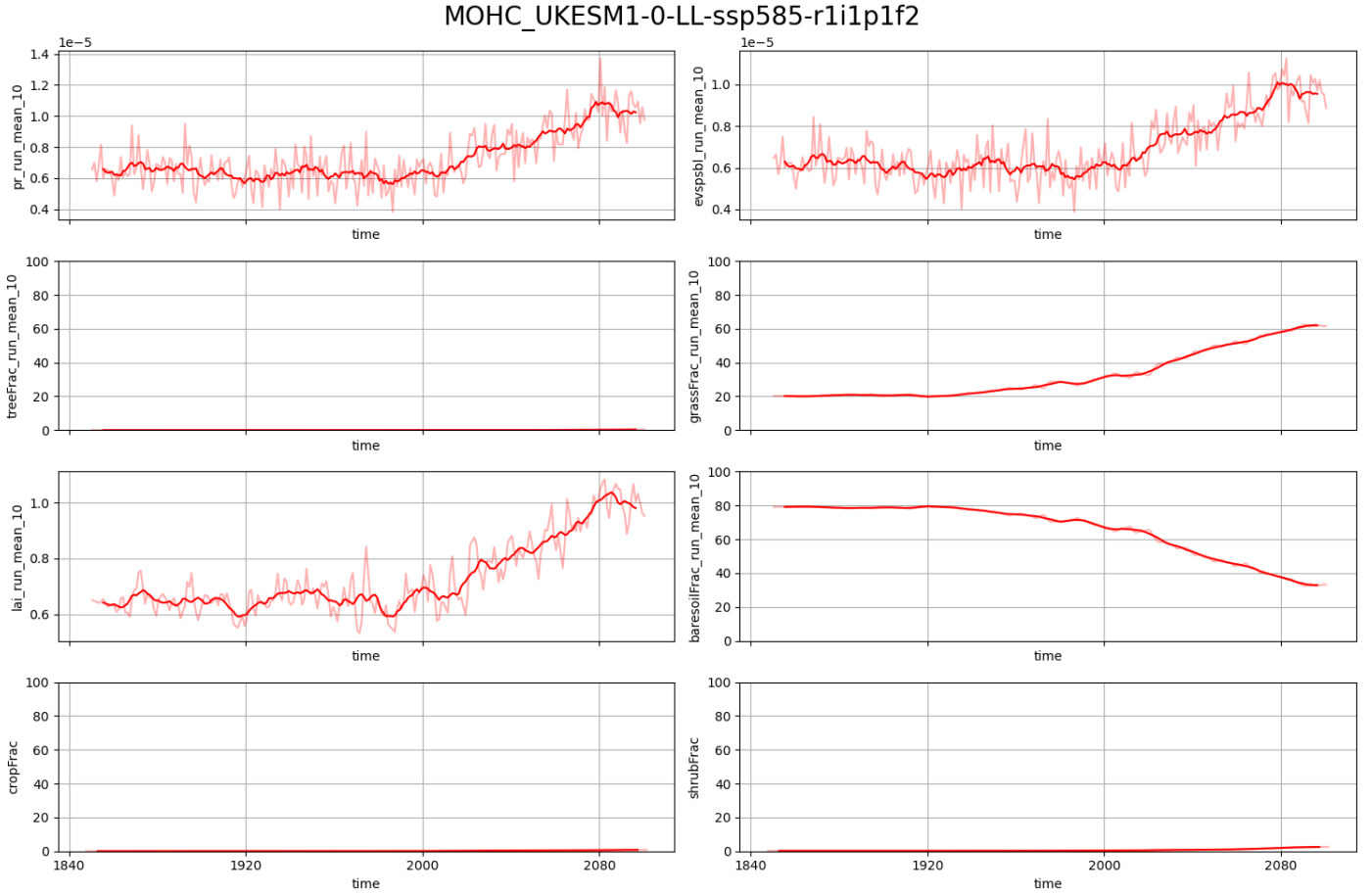


Figure 26: Time series of the case: gradual decrease in bare-soil fraction

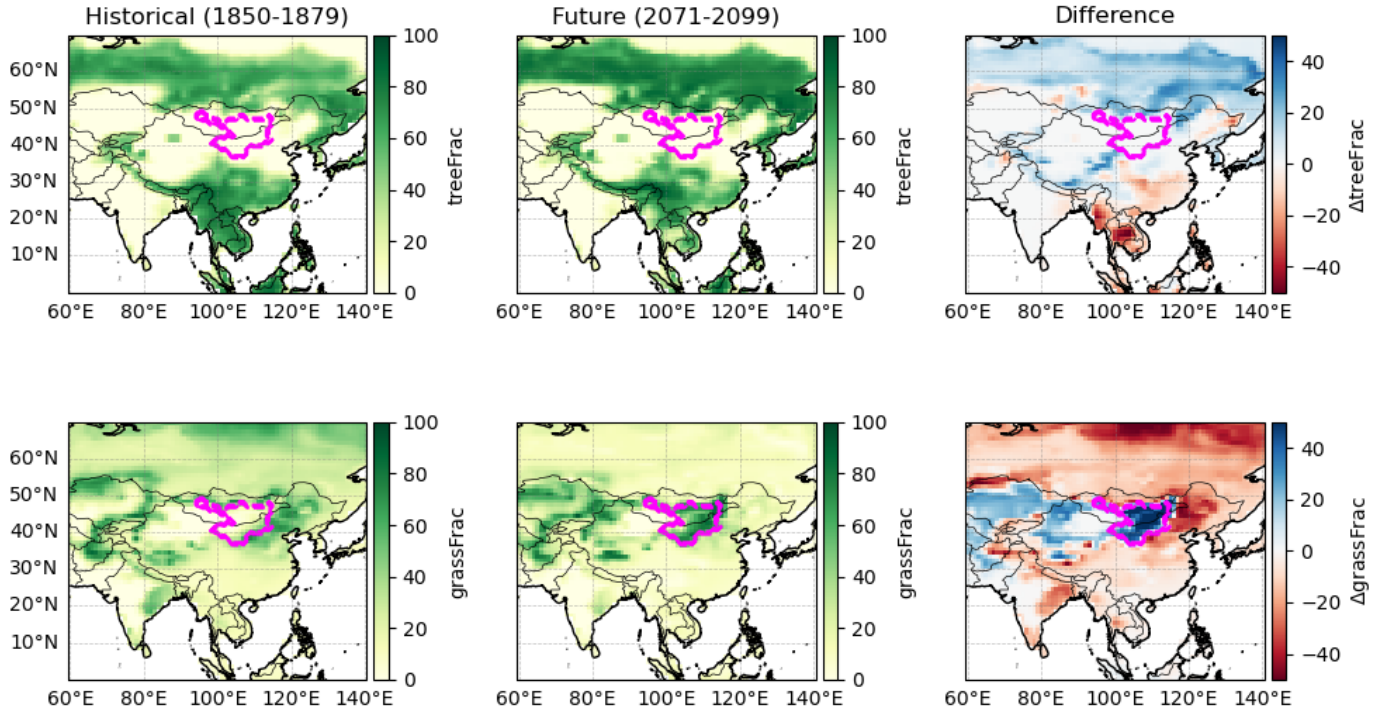


Figure 27: Spatial maps of the case: gradual decrease in bare-soil fraction

3.3.2 UKESM1-0-LL (r4i1p1f2) – Gradual increase in tree fraction

The SSP5-8.5 scenario of this model projects a gradual increase in tree fraction in central China, from 20% during 1850–2060 to around 80% by 2300 (Figs. 28, 29). The model also projects a decrease in grass fraction over the region of around 30%. Growth of land-use-related vegetation covers in the region (i.e., crop and pasture) is small (6%) and only continues until 2100. Precipitation flux over the region also increases by $1.5 \text{ kg m}^{-2} \text{ s}^{-1}$, which may create favorable conditions for tree-area expansion.

MOHC_UKESM1-0-LL-ssp585-r4i1p1f2

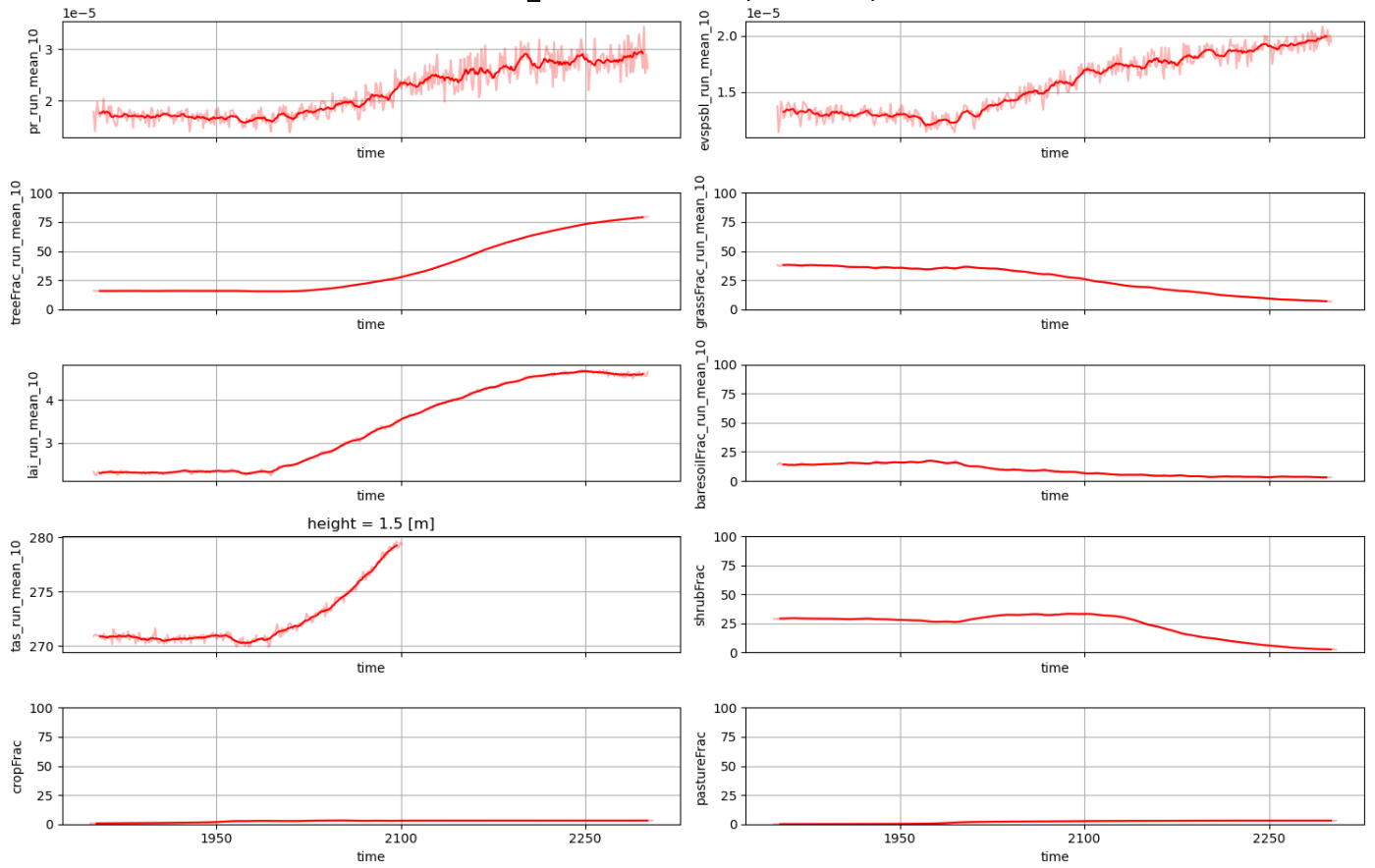


Figure 28: Time series of the case: gradual increase in tree fraction

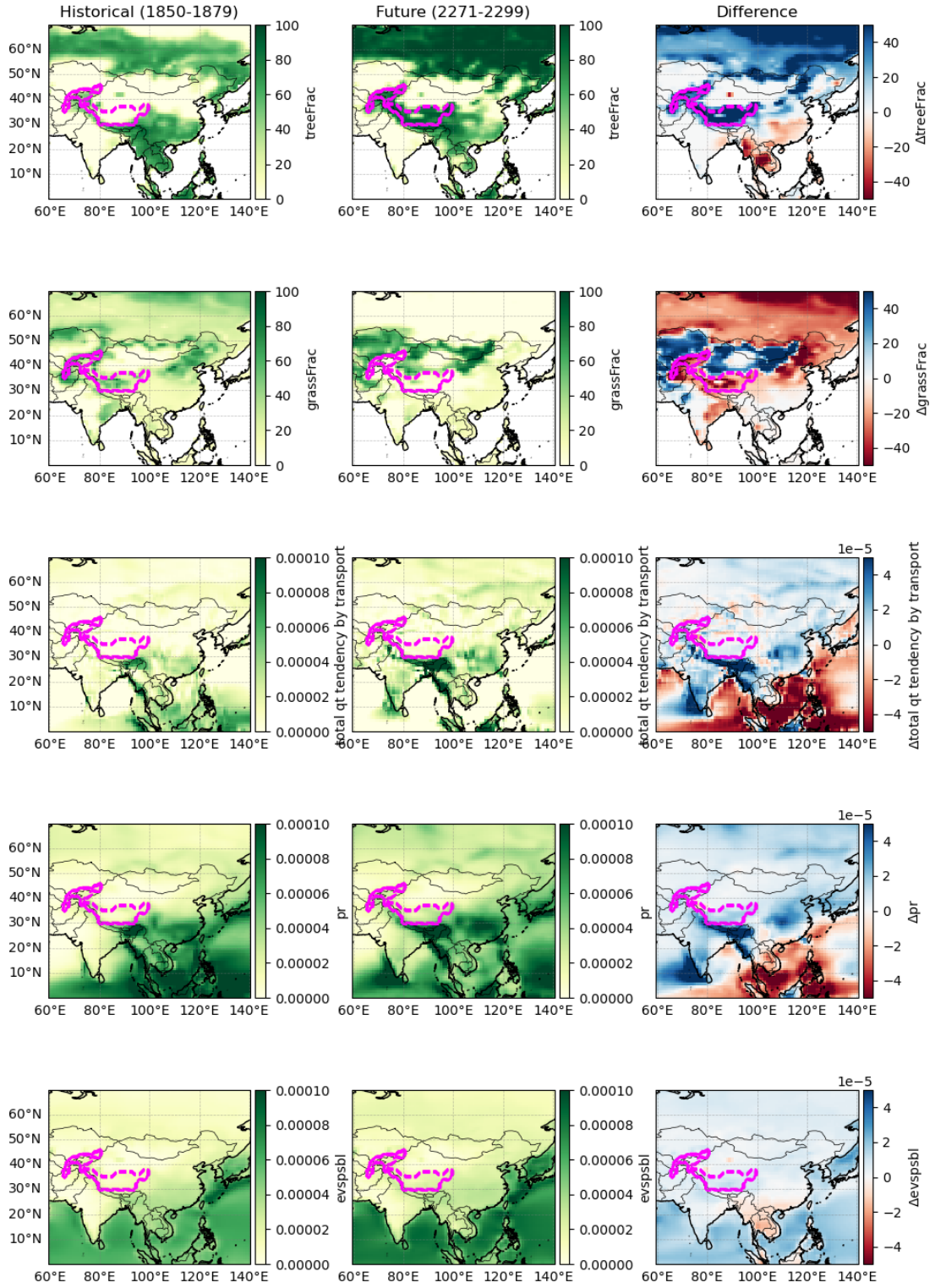


Figure 29: Spatial maps of the case: gradual increase in tree fraction

Moisture budget analysis over the region shows that around 75% of total precipitation can be explained by evapotranspiration (Fig. 30). Nevertheless, part of the precipitation increase is also due to enhanced horizontal moisture

transport into the region. Thus, it is possible that, with increasing precipitation, trees expand, which in turn drives more evapotranspiration, creating a feedback that further increases precipitation over the region.

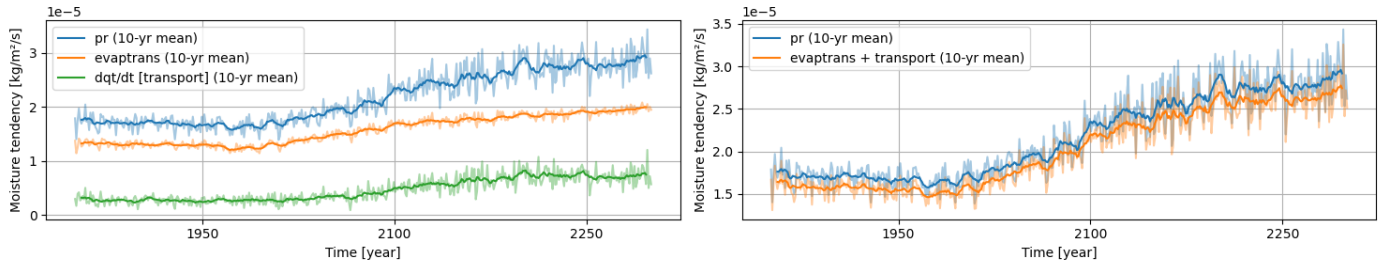


Figure 30: Time series of the precipitation-related metrics for the case: gradual increase in tree fraction

3.3.3 Increased leaf-area index in six non-dynamic vegetation climate models over China

The SSP5-8.5 scenario of the climate models CanESM5, NorESM2-LM, CanESM5-1, and CanESM5-CanOE in southeastern China, and CESM2 and CESM2-WACCM in southern China, projects a gradual increase in leaf-area index. Most of these non-dynamic vegetation models also show an increase in precipitation over the regions. Nevertheless, some models, such as NorESM2-LM and CanESM5-CanOE (see its time series and spatial maps in Figs. 31, 32), do not display a clear pattern in precipitation and evapotranspiration over the region, leaving the underlying mechanisms in these models unclear.

CCCma_CanESM5-CanOE-ssp585-r1i1p2f1

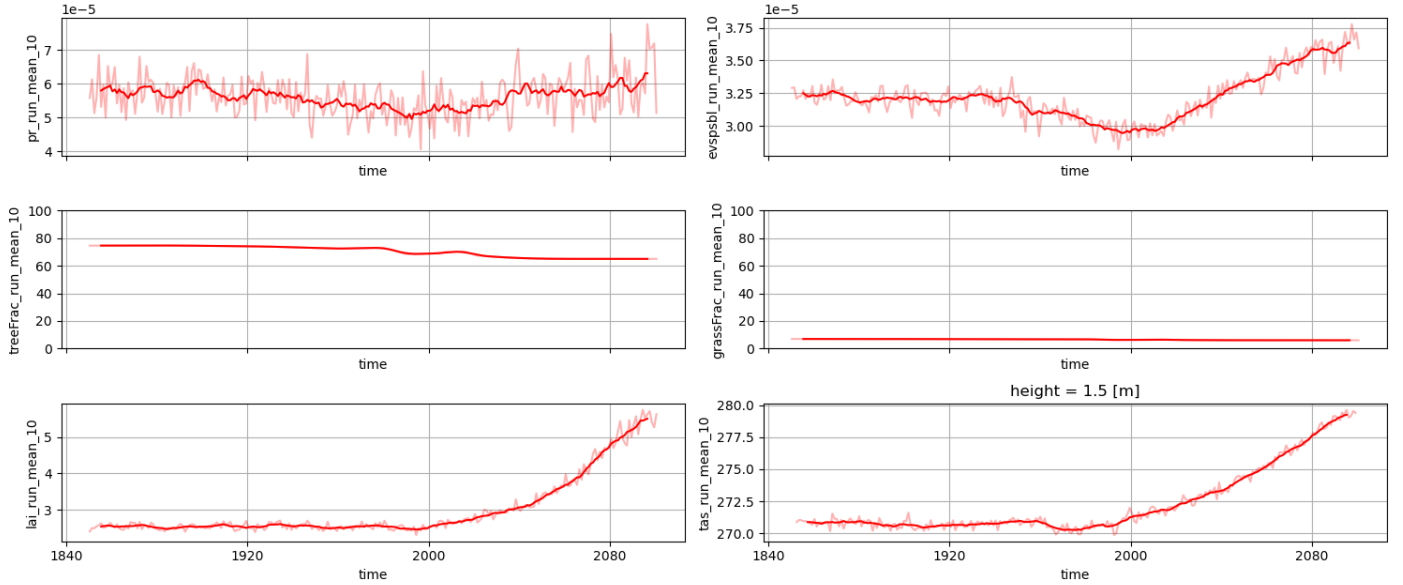


Figure 31: Time series of the case: gradual increase in leaf-area index

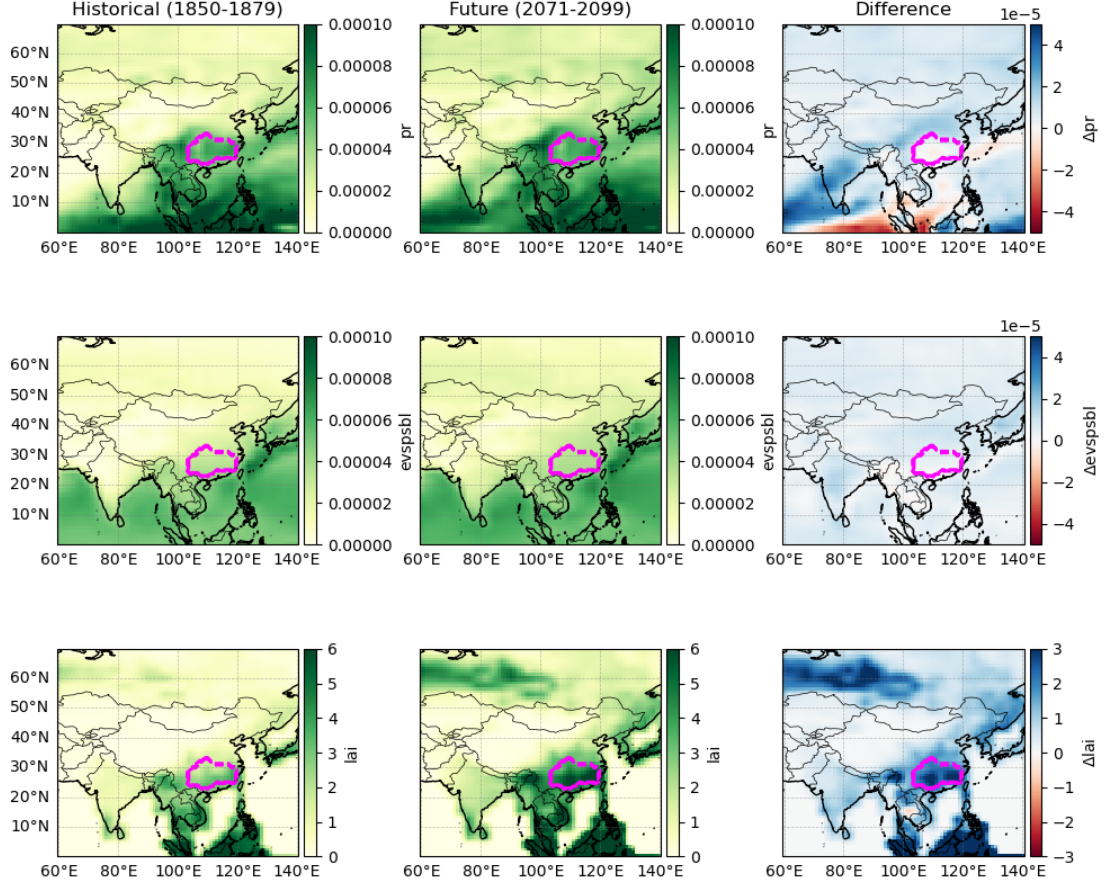


Figure 32: Spatial maps of the case: gradual increase in leaf-area index

3.3.4 Increased leaf-area index in two non-dynamic vegetation climate models over the USA

The SSP5-8.5 scenario of the climate models CanESM5-1 (see its related time series and spatial maps in Figs. 33, 34) and CanESM5-CanOE projects an increase in leaf-area index over the eastern part of the USA. This increase is consistent with the observed rise in precipitation and evapotranspiration over the region.

CCCma_CanESM5-1-ssp585-r4i1p1f1

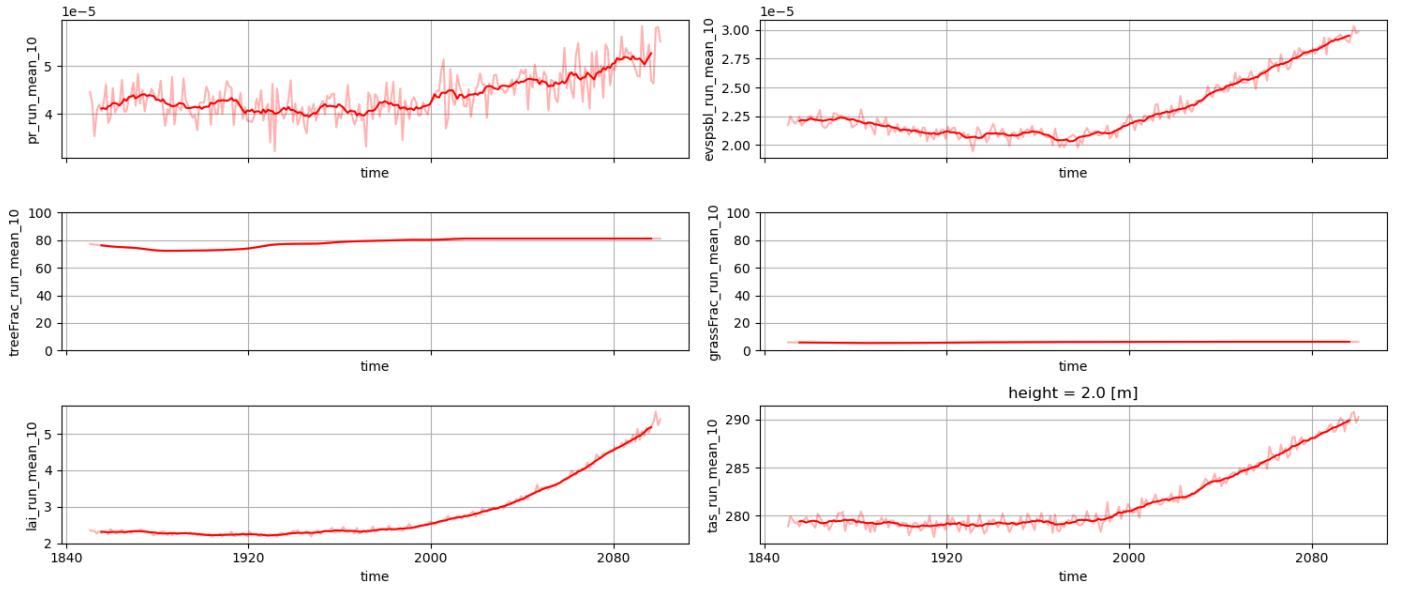


Figure 33: Time series of the case: gradual increase in leaf-area index

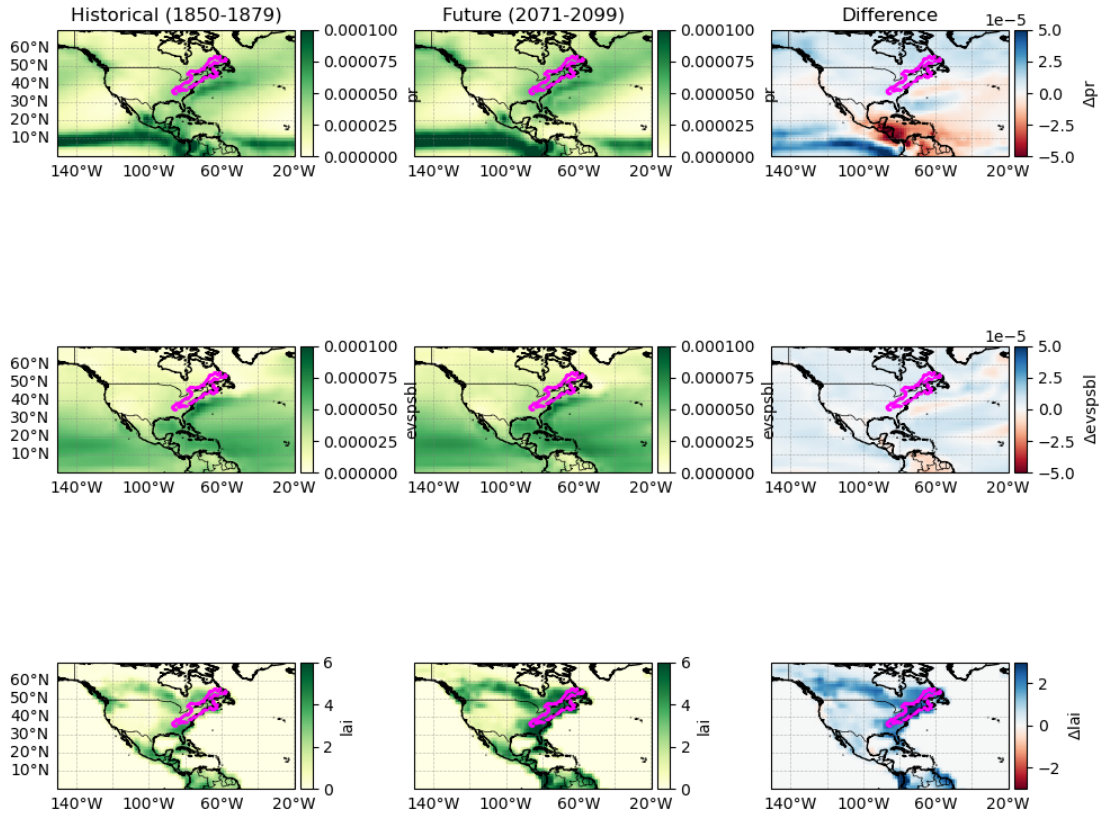


Figure 34: Spatial maps of the case: gradual increase in leaf-area index

3.4 Arctic zone

In this section, we present the dramatic shifts in the Arctic zone, including eighteen cases related to permafrost collapse, two cases associated with tree-area expansion in dynamic vegetation models, five cases with increased leaf-area index in non-dynamic vegetation models, and, finally, two cases with decreased snow-area fraction over the region.

3.4.1 Thirteen cases with gradual and abrupt decrease of total soil moisture

The climate models CNRM-CM6-1, CNRM-CM6-1-HR, and CNRM-ESM2-1 project a gradual decrease in total soil water content for seven cases over the far north of Asia and North America. While all these models show such dramatic shifts in land permafrost under SSP5-8.5 scenarios, CNRM-CM6-1 and CNRM-CM6-1-HR display this transition consistently across all the other scenarios, including SSP1-2.6, SSP2-4.5, and SSP3-7.0. Moreover, similar but abrupt transitions in total soil moisture over the same regions are projected for five cases in northern Asia and North America by the climate models ACCESS-ESM1-5, ACCESS-CM2, E3SM-1-1, and UKESM1-0-LL (SSP3-7.0 and SSP5-8.5)

In all the models mentioned above, the significant decrease in total soil moisture is primarily driven by a strong reduction in frozen soil water content, which is usually about twice the decrease in total soil moisture (see an example model in Fig. 35). This implies that roughly half of the water remains in the soil in liquid form, while the rest drains out. The decrease in frozen soil water is driven by rising temperatures across the region, and it generally occurs when the number of months with a mean temperature above zero exceeds five per year in all the models mentioned. Notably, in cases with abrupt transitions, the frozen water content completely vanishes within a few decades after the abrupt transition begins.

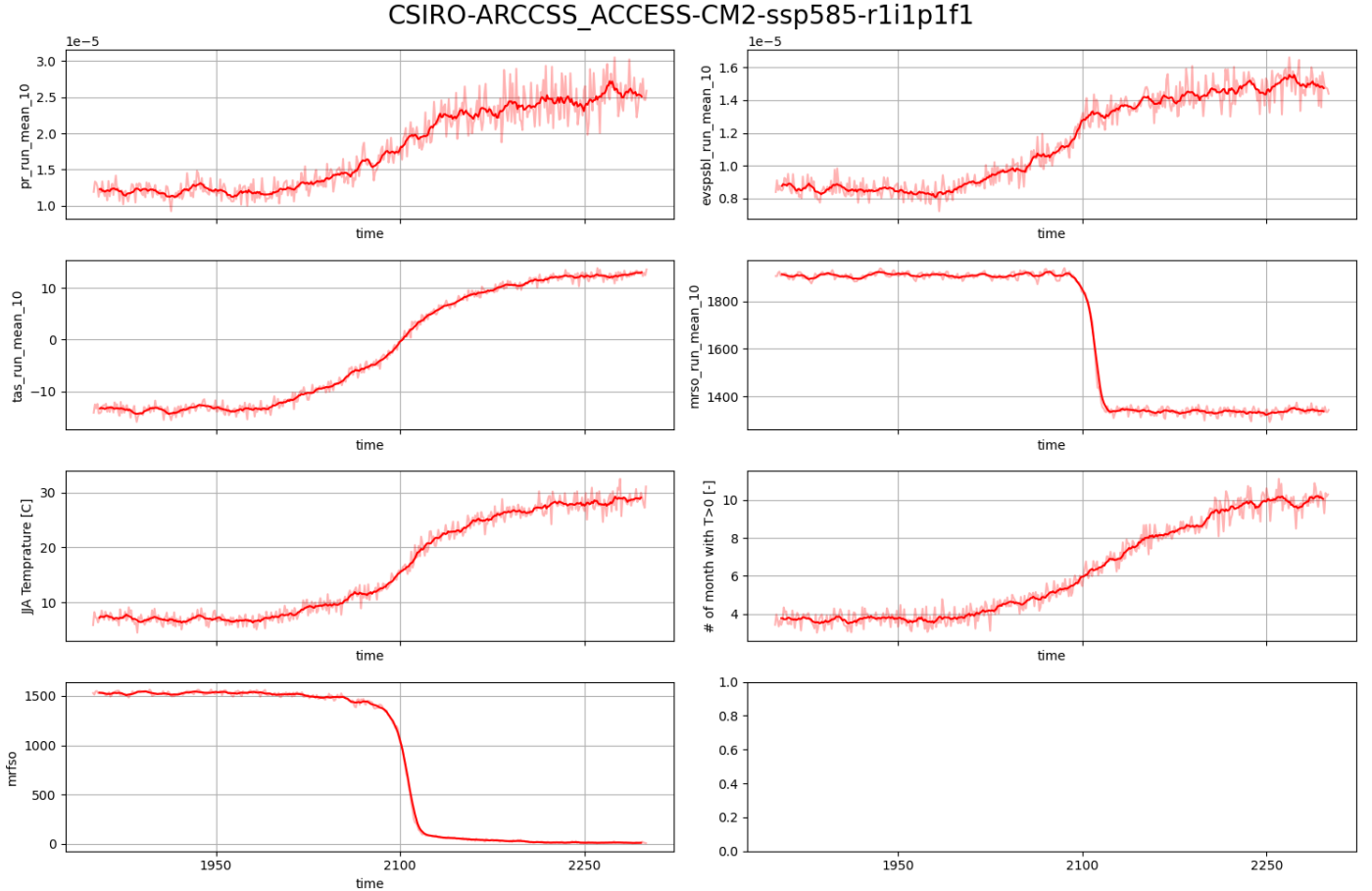


Figure 35: Time series of an example model for the case: abrupt decrease in total soil water content over Arctic zone

3.4.2 Five cases with gradual increase in total soil moisture

Climate models TaiESM1, NorESM2-LM, and NorESM2-MM, for five cases, project a gradual increase in total mass water content in soil (see an example model (NorESM2-LM) in Fig. 36). After an investigation, we realized that these models

only include the liquid phase in their outputs of total mass water content of soil while the frozen phase is not included. Analyzing the frozen water content separately for these cases, we still find a gradual decrease in frozen water content for these cases but unlike the cases of the previous section, the melted water does not drain out form the soil layer and thus remain in the soil.

These cases also exhibit an increase in region-mean precipitation and a decrease in snow-area fraction, both of which provide additional sources of liquid water for soil moisture over the region.

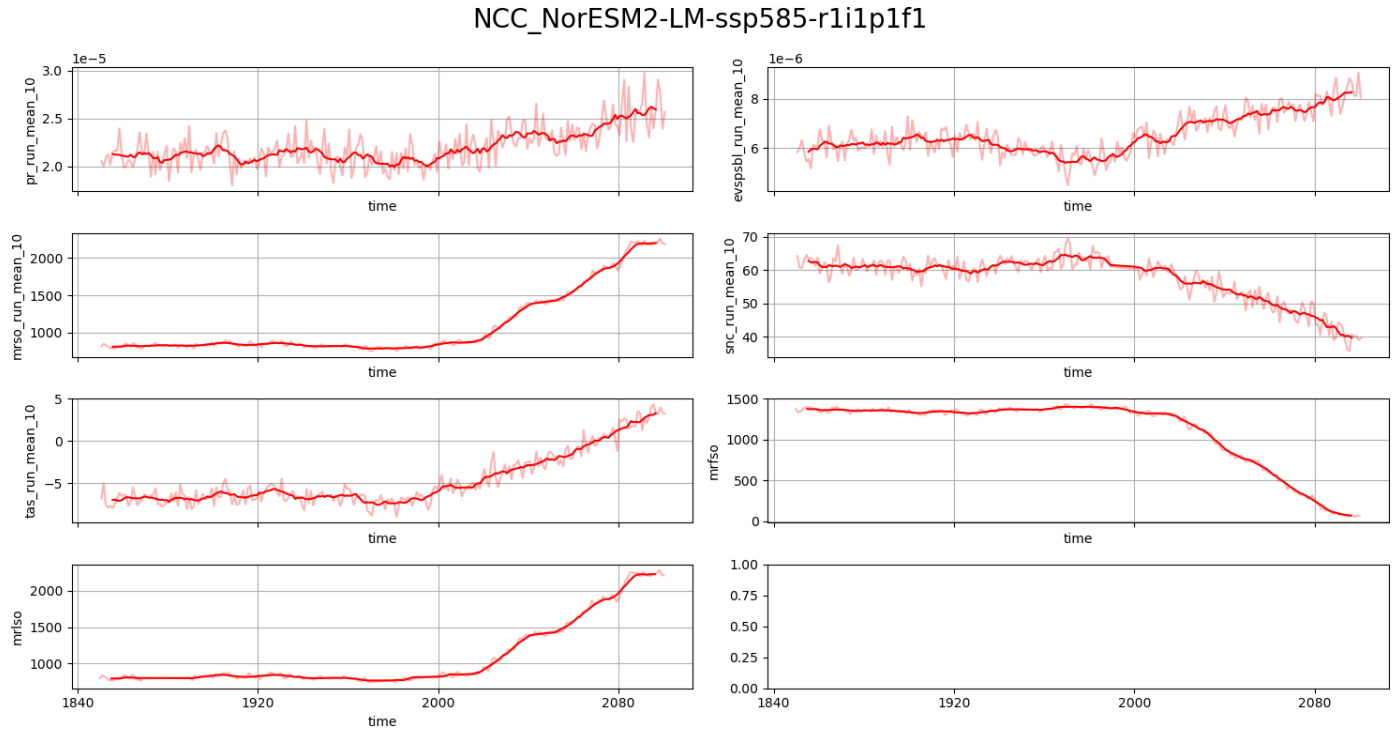


Figure 36: Time series of an example model for the case: gradual increase in total soil water content over Arctic zone

3.4.3 UKESM1-0-LL (r4i1p1f2) – Two cases with a gradual increase in tree fraction

The SSP5-8.5 scenario of this model projects a gradual, significant increase in tree fraction from 20% in the 1850-2050 period to 80% by 2300 (e.g., Fig. 37 for the far north of Asia). In the same region and period, grass-area fraction completely vanishes (by 40%) and bare-soil area fraction also decreases by 20% to nearly zero. This member also projects significant reductions in frozen soil water content, leading to the majority of water in the soil being only in the liquid phase. Such conditions, with increased temperature and precipitation, create favorable conditions for tree expansion over the region. In response, leaf-area index and evapotranspiration increase over the region.

MOHC_UKESM1-0-LL-ssp585-r4i1p1f2

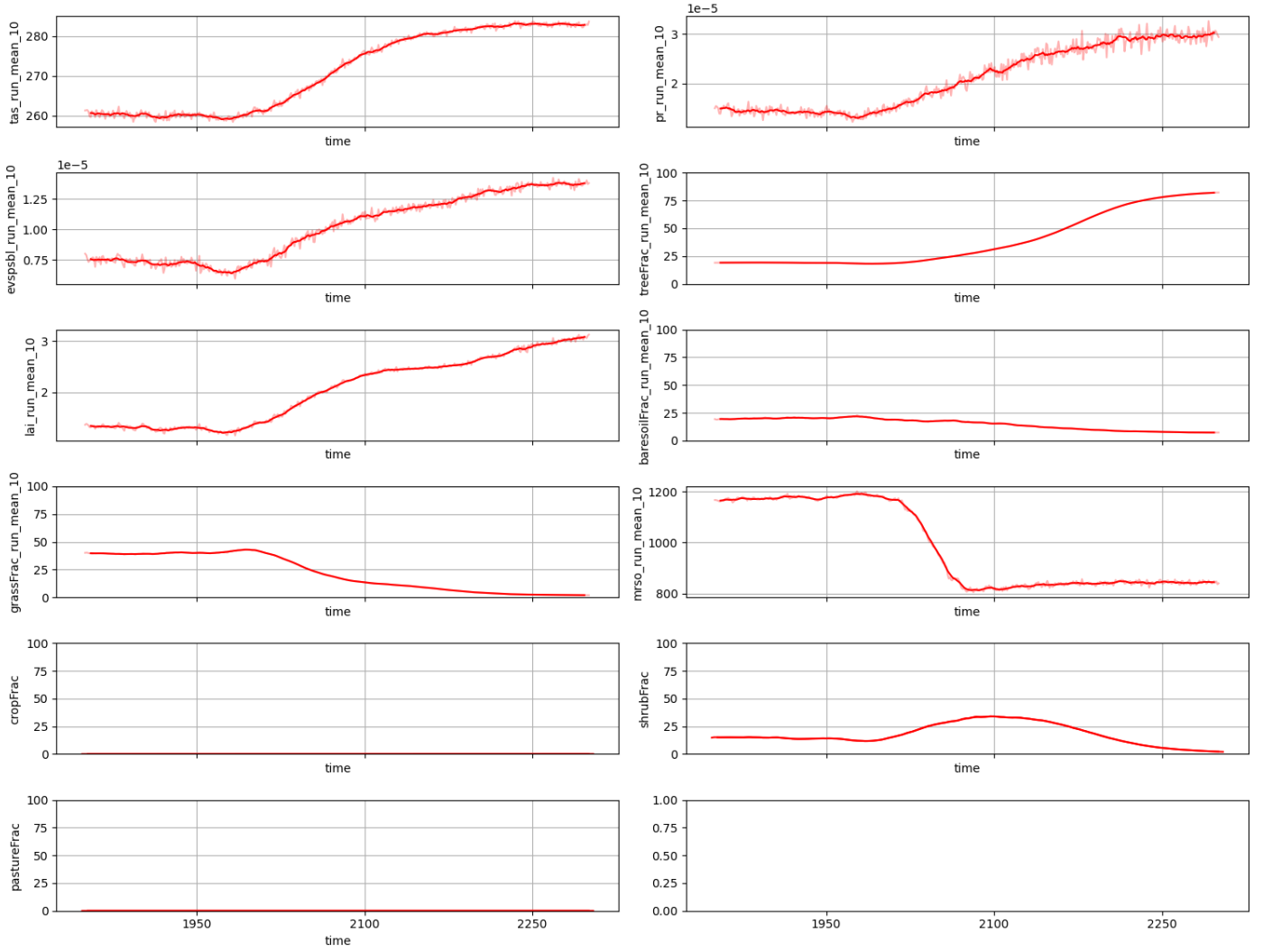


Figure 37: Time series of one of the cases for the case: gradual increase in tree fraction over Arctic zone

3.4.4 Increased leaf-area index in three non-dynamic vegetation models over Arctic zone

The SSP5-8.5 scenario of the models CanESM5-CanOE (1 case), CanESM5-1 (1 case), and CanESM5 (2 cases; see its time series for one example case in Fig. 38) projects a gradual increase in leaf-area index over the Arctic zone in northern North America and Asia. In all cases, this increase is consistent with higher precipitation and evapotranspiration over the regions.

CCCma_CanESM5-ssp585-r1i1p1f1

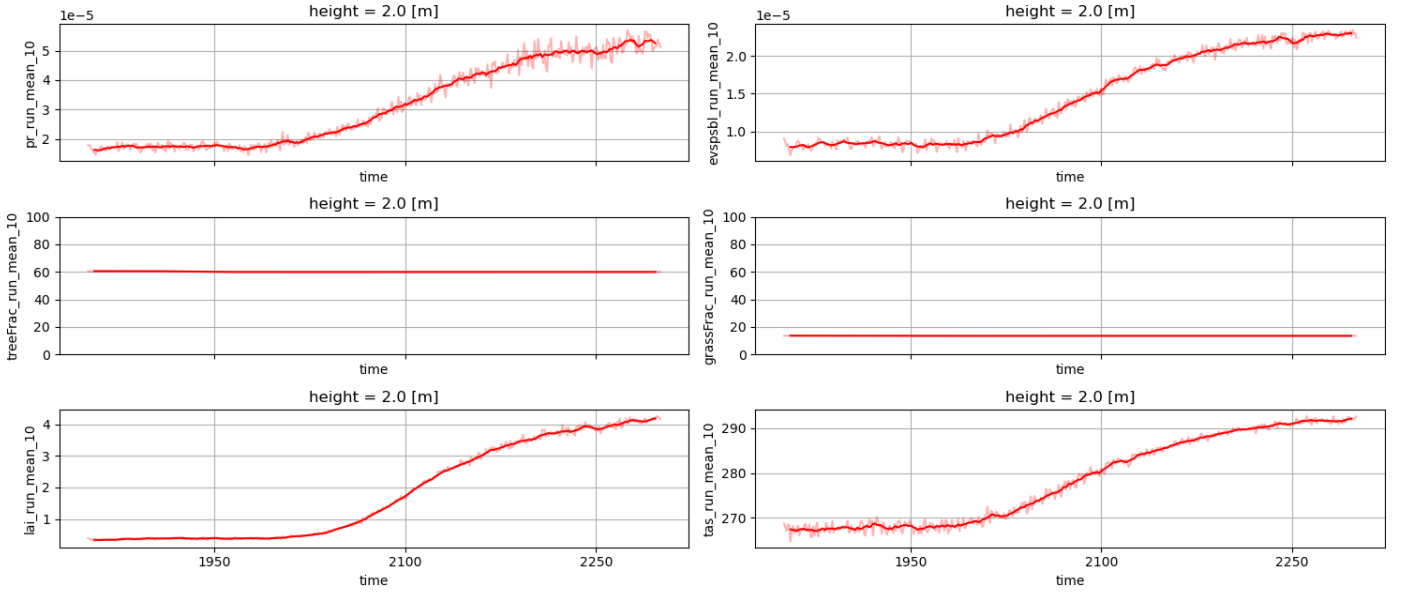


Figure 38: Time series of one of the cases for the case: gradual increase in leaf-area index over Arctic zone

3.4.5 NorESM2-MM (r1i1p1f1) and GISS-E2-1-G (r5i1p1f2): gradual decrease in snow-area fraction

The SSP5-8.5 scenario of these two models projects a gradual decrease in snow-area fraction, from full coverage in the 1850–2000 period to around 45% yearly mean over the far northeast of North America (Fig. 39). The gradual decrease appears to begin immediately after the summer-averaged (JJA) temperature crosses the critical threshold of 0 degrees, which coincides with the first occurrence of a month with above-zero monthly-mean temperature within a year. The snow-area fraction averaged over the summer completely vanishes by 2200 in the region.

NASA-GISS_GISS-E2-1-G-ssp585-r5i1p1f2

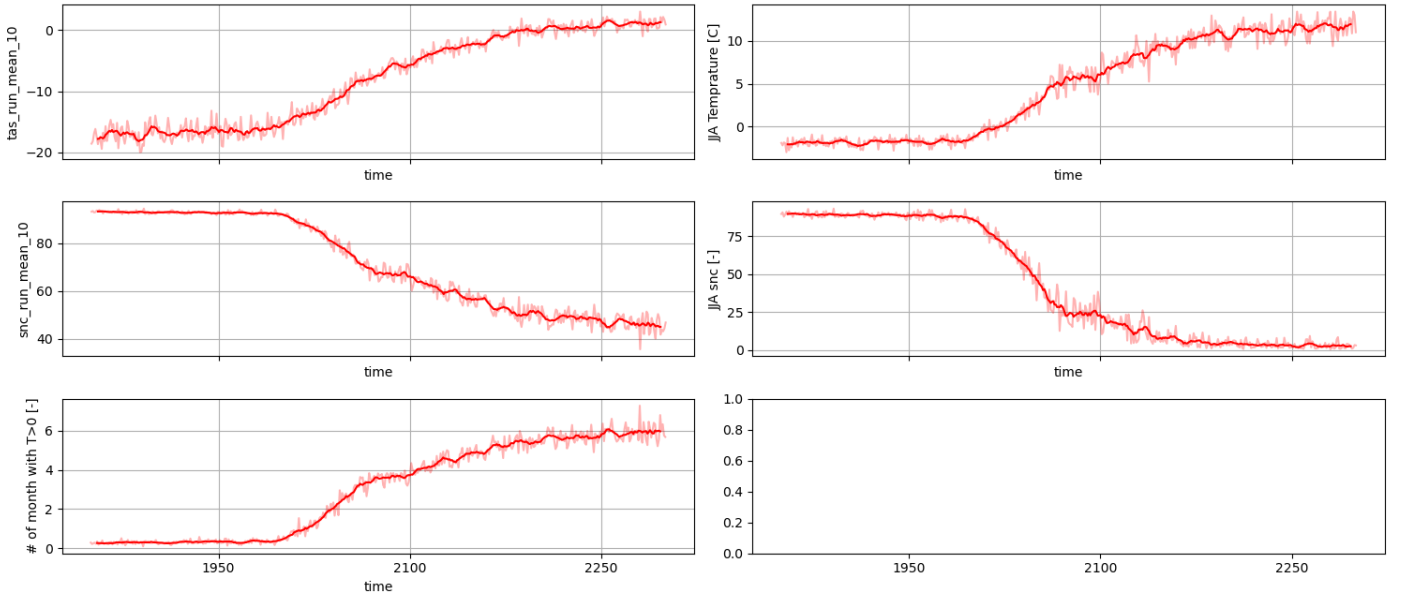


Figure 39: Time series of one of the cases for the case: gradual decrease in snow-area fraction

4 Summary & Conclusion